WATER LEAKAGE & WASTAGE REPORTING MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

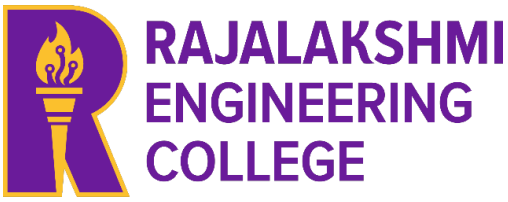
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in partial fulfilment of the End Semester Examination for the course

CB23F35 –
LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING
COMPUTER SCIENCE AND ENGINEERING



(An Autonomous Institution, Affiliated to Anna University, Chennai)
Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

November, 2026

BONAFIDE CERTIFICATE

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Certified that this project report titled "Water Leakage & Wastage Reporting Management System" is the bonafide work of T KAUSHIKA (240701239), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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I, T KAUSHIKA (240701239), hereby declare that the project report titled “Water Leakage & Wastage Reporting Management System”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report, including its forms, reports, dashboard pages, Deluge scripts, workflows and notifications, was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as the sentiment-analysis field), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: 10 November 2026

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ACKNOWLEDGEMENT

I express my sincere gratitude to our Chairman Thiru. S. Meganathan, Chairperson Dr. (Mrs.) Thangam Meganathan and Vice-Chairman Mr. Abhay Meganathan S for providing the infrastructure and facilities that made this work possible.

I am grateful to our Principal, Dr. S. N. Murugesan, Rajalakshmi Engineering College, for the constant encouragement extended to students to take up industry-relevant learning.

I thank Dr. Manoranjini J, Head of the Department, Department of Computer Science and Engineering, for the guidance and support provided throughout the course.

I owe my sincere thanks to my course faculty, Mrs. Aswana Lal, for the valuable guidance, reviews and feedback that shaped this application and report.

I also acknowledge Zoho Corporation for providing access to the Zoho Creator platform and its learning resources, and I thank my family, friends and classmates for their unwavering support.

T KAUSHIKA

ABSTRACT

Water supply systems lose a large share of their water through leakage, bursts, overflowing tanks and wastage at public taps. The complaints about these losses, the repairs that follow and the assets that need inspection are usually kept in registers, messages and spreadsheets, so the administration cannot easily see which complaints are open, which are overdue or where water is being lost. The Water Leakage & Wastage Reporting Management System is a low-code application built on Zoho Creator that brings these records into one system. The application has eleven forms: Area/Locality Master, Citizen Registration, Water Issue Report, Repair Resolution, Technician, Maintenance Team, Water Asset Location, Materials Used, Inspection Checklist, Evidence Subform and Feedback. Three grid subforms link photos, materials and checklists to their parent records. Six form workflows and one scheduled workflow written in Deluge move each complaint through submission, assignment, repair and supervisor verification, and send e-mail notifications to citizens, technicians and supervisors. The application provides 27 reports, four role-based dashboards with KPI panels and charts, role-based profiles for administrators, supervisors, technicians and citizens, and a sentiment-analysis field on citizen feedback. Each role sees only the forms, reports and dashboards it needs, enforced through profiles and permissions defined for administrators, supervisors, technicians and citizens. The planned validation consists of 22 test cases, and a review of the exported application recorded eight findings with proposed fixes. The application replaces scattered registers with validated, linked and tracked records, and gives management a live view of water complaints, repairs and assets, so that leakages can be attended to sooner and water losses can be reduced.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Water Management; Complaint Tracking; Dashboard

PROJECT AT A GLANCE

Table G.1: Project at a Glance

Attribute	Details
Project Title	Water Leakage & Wastage Reporting Management System
Application Domain	Municipal water management – leakage, wastage and public water complaints
Student Name & Register No.	T Kaushika – 240701239
Zoho Creator Edition	Zoho Creator (India data centre, creator.zoho.in)
Application Name	Water Leakage&Wastage Reporting Management System (icon WLR)
Published / Live URL	https://water-complaint-portal.zohocreatorportal.in
Demo Credentials (per role)	End customer can directly access the published component. Admin, citizen, techician and supervisor user access production deployment using zoho creator account
Number of Forms / Subforms	11 / 3 grids
Number of Reports	27 (26 list, grouped and criteria-based; one pivot chart)
Number of Pages / Dashboards	4 (Admin, Supervisor, Technician, Citizen)
Number of Workflows	Form: 6; Schedule: 1; Approval: 0; Blueprint: 0; Batch: 1
Deluge Functions	0 custom functions; Deluge scripts in workflows
Integrations	E-mail notifications (sendmail); AI sentiment-analysis field
Roles & Profiles Defined	Administrator, Admin, Supervisor, Technician, Citizen, Read, Write, Developer, Customer; hierarchy Admin > Supervisor > Technician, Admin > Citizen
Environments Used	Development → Stage → Production
Date of Production Deployment	06 Nov 2026
Configuration Export	Version 1.0, generated 27 October 2026

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LIST OF ABBREVIATIONS

Table X.1: Abbreviations

Abbreviation	Expansion
AI	Artificial Intelligence
CO	Course Outcome
DS	Deluge Script
FR	Functional Requirement
INR	Indian Rupee
IST	Indian Standard Time
KPI	Key Performance Indicator
NFR	Non-Functional Requirement
PDF	Portable Document Format
RBAC	Role-Based Access Control
UAT	User Acceptance Testing
UI	User Interface

CHAPTER 1
INTRODUCTION

1.1 Background

Water is supplied to a city through pipelines, public taps, storage tanks, pumps and meters, and a large share of it is lost through leakage, bursts, overflowing tanks and wastage at public taps. Each loss is reported by citizens, noticed by field staff or discovered during inspection, and then has to be recorded, assigned to a repair team, repaired, verified and closed. In many local bodies this chain still runs on paper registers, verbal complaints and messaging groups, so it is hard to know which complaints are open, which are overdue and where water is being lost most often.

This project replaces that scattered record-keeping with a single Zoho Creator application, the Water Leakage & Wastage Reporting Management System. The application stores citizen registrations, water issue reports, water assets, maintenance teams, technicians, repair records, materials, inspection checklists and citizen feedback in one database. It routes each complaint through assignment, repair and supervisor verification, notifies the people involved by e-mail, and presents the state of the work on role-specific dashboards and reports.

1.2 Problem Statement

Water complaints and repair records are kept in disconnected registers and messages, so the administration cannot easily see the number of open complaints, their priority, the team responsible, the repair progress or the water lost in each locality. A system is needed that lets citizens report problems once, lets supervisors assign and verify work, lets technicians record what they did, and gives decision makers a live view without manual summaries. The main problems are:

- Complaint tracking: reports arrive informally and are rarely followed to closure.
- No ownership: a complaint has no assigned team, technician, priority or expected resolution date.
- Repair records: work done, materials used, inspection results and water loss are not recorded in a consistent form.

- Asset knowledge: pipelines, tanks, pumps, valves and meters have no register with inspection dates and responsible teams.
- Citizen voice: feedback after resolution is not captured, so service quality cannot be measured.
- Slow insight: answering “how many high-priority complaints are overdue in one district?” needs several files and manual counting.

1.3 Objectives

- To design a connected data model and forms for citizens, areas and localities, water issues, water assets, maintenance teams, technicians, repairs, materials, inspection checklists and feedback in Zoho Creator.
- To route each water complaint through submission, assignment, repair, verification and resolution using form workflows that update the complaint status automatically.
- To notify citizens, technicians and supervisors by e-mail at each important step.
- To provide role-based dashboards and reports that summarise open, overdue, high-priority and resolved complaints, repair performance and water asset status.
- To send a daily overdue-complaint alert using a scheduled workflow.
- To analyse citizen feedback sentiment through the platform’s AI field, and to restrict each user profile to the data it needs.

1.4 Scope of the Project

In scope. The application contains eleven forms (Area/Locality Master, Citizen Registration, Water Issue Report, Repair Resolution, Technician, Maintenance Team, Water Asset Location, Materials Used, Inspection Checklist, Evidence Subform and Feedback), three grid subforms, 27 reports, four role-specific dashboard pages, six form workflows, one scheduled workflow and one batch workflow, e-mail notifications, a sentiment-analysis field on feedback comments, role and profile based access, and phone and tablet layouts.

Out of scope. The application does not read water meters or leakage sensors automatically; all readings are entered by users. It does not calculate water-loss volumes

from flow data, does not integrate with a payment or billing system, does not send SMS messages, and does not use a CRM or external REST API in this version. These items are listed as future enhancements in Chapter 9.

1.5 Justification for a Low-Code Approach

A water complaint and repair tracker is a typical internal business application: a set of related records, a status workflow, notifications, reports and dashboards. Zoho Creator provides these building blocks on a hosted platform, including forms with lookups, grids, form workflows, a scheduler, reports, a page builder, Deluge scripting, user profiles and a mobile client. Hosting, authentication, the database and the mobile app did not have to be built, so the effort could go into the parts that matter for the service: the status lifecycle, the role separation and the dashboards.

Low-code platforms let developers and domain experts build applications through visual modelling with limited hand-written code. This suited a project built by one student within one semester. Where visual tools were not enough, short Deluge scripts were written in form workflows for status changes, e-mail notifications and the overdue alert.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements. The appendices contain the feature coverage checklist, course outcome mapping, the Deluge source code, a user manual and the evaluation sheets.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The existing process for handling water complaints, as followed in most local bodies and campuses, works as follows:

1. Citizens report leakages or water problems by visiting an office, by phone or through a local representative.

2. The complaint is written in a register, often without the location, priority or contact details needed to act on it.

3. A supervisor decides which team should attend, usually by telephone, and the decision is not recorded.

4. Technicians repair the problem and report back verbally; materials used and water lost are rarely written down.

5. Inspections of pipelines, tanks and pumps follow a separate schedule kept on paper or in spreadsheets.

6. At the end of a month, an administrator compiles the scattered figures manually into a report for management.

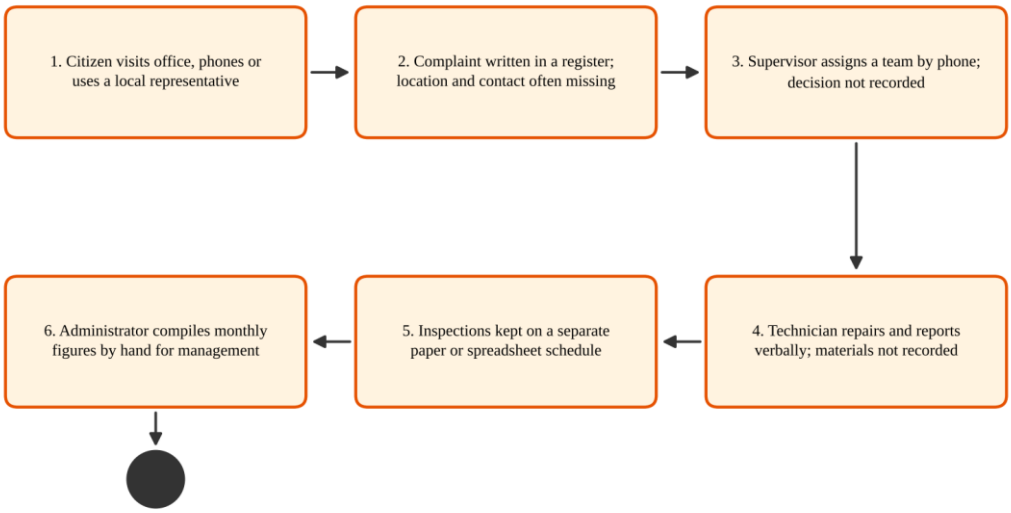


Figure 2.1: As-Is Process Flow of Water Complaint Handling

2.2 Limitations of the Existing System

- Manual and repeated data entry: the same details are written in registers and then re-typed into reports.
- No single view of open, overdue and resolved complaints across localities.
- No ownership or deadline for complaints, so overdue work is not noticed.
- Repair work, materials and inspection results are not linked to the complaint they fix.
- Water assets have no inspection schedule or responsible team.
- Citizen feedback after resolution is not collected, so service quality is unknown.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Several weeks to months for database, back-end, front-end and authentication	Built within one semester by one student using forms, workflows and pages
Skills required	Full-stack development, database design, front-end charting, DevOps	Zoho Creator configuration and Deluge scripting
Hosting	Servers must be provisioned, secured and maintained	Hosted by Zoho; no server administration
Notifications	E-mail service must be integrated by hand	Built-in sendmail task in workflows
Mobile app	Separate Android/iOS app must be built	Zoho Creator mobile app; phone and tablet layouts defined
Security & access	Authentication and roles implemented by hand	Built-in profiles, roles and personal-data flags
Customisation limits	Unlimited	Limited to platform components; complex logic needs scripting

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title	Key Findings	Relevance to this Project
1	A. Sahay, A. Indamutsa, D. Di Ruscio and A. Pierantonio, 2020 [1]	Supporting the understanding and comparison of low-code development platforms	Feature-based comparison of low-code platforms covering data modelling, UI, logic and deployment	Vocabulary used to map this project's features
2	A. C. Bock and U. Frank, 2021 [2]	Low-Code Platform	Characterises low-code platforms and their effect on productivity and agility	Supports building an institutional application with a single developer
3	D. Di Ruscio et al., 2022 [3]	Low-code development and model-driven engineering	Compares low-code with model-driven engineering	Frames forms, lookups and workflows as models from which the application runs
4	Zoho Corporation [4], [5]	Zoho Creator Help and Deluge Reference	Official documentation of forms, workflows, reports, pages, schedules and Deluge	Primary technical reference for Chapter 5

2.5 Emerging Trends in Low-Code Development

Two trends shaped this project. The first is citizen development combined with pro-code extension: platforms let non-specialists build most of an application visually while developers add scripts where needed [2], [3]. Here, forms, reports and dashboards are configured visually, while status changes and e-mail notifications are written in Deluge. The second is AI assistance inside business applications. Zoho Creator offers AI fields, and this project uses one such field to analyse the sentiment of citizen feedback comments.

2.6 Proposed System

The Water Leakage & Wastage Reporting Management System addresses each limitation of the existing system. Citizens register once and report issues with a type, severity, location, photos and a landmark. Each new report starts with the status "submitted", is confirmed to the citizen and the administrator by e-mail, and is then assigned to a technician. The technician records repair times, cost, water-loss estimate,

materials and inspection checklist results. When the repair is completed the supervisor verifies it: an approved repair resolves the complaint and informs the citizen, while a repair needing rework returns the complaint to in progress. Citizens then submit feedback, and low ratings or requests for follow-up are sent to the administrator. Dashboards show each role what it needs to see.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Application owner with full access	Maintains master data (areas, teams, technicians, assets), reviews all dashboards and reports, receives feedback follow-up alerts
Supervisor	Manages complaints and verifies repairs	Reviews and assigns complaints, monitors repair progress, verifies completed repairs, maintains citizen and asset records
Technician	Field staff who repair water issues	Views assigned complaints, records repair details, materials, inspection results and photos
Citizen	Resident who reports problems	Registers, reports water issues, tracks own complaints and submits feedback after resolution
External e-mail recipients	Administrator, supervisor, technician and citizen mailboxes	Receive confirmation, assignment, verification and overdue notifications

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall maintain area and locality masters for the 38 districts of Tamil Nadu and register citizens against them.	Administrator, Citizen	High
FR-02	The system shall let a citizen report a water issue with type, severity, priority, location, landmark, description and up to ten photos.	Citizen	High
FR-03	The system shall confirm each new complaint to the citizen and the administrator by e-mail.	System	High
FR-04	The system shall let a supervisor assign a maintenance team and technician to a complaint and set the status to Assigned.	Supervisor	High
FR-05	The system shall record repair work by technicians, including start and end times, cost, water-loss estimate, materials used and inspection checklist results.	Technician	High
FR-06	The system shall update the complaint status from repair progress and from supervisor verification (Approved: Resolved; Rework Required: In Progress).	System, Supervisor	High
FR-07	The system shall maintain a register of water assets with type, status, inspection dates and responsible team.	Administrator, Supervisor	Medium

FR ID	Requirement	Role(s)	Priority
FR-08	The system shall collect citizen feedback with a rating, comments, follow-up flag and sentiment, and alert the administrator for low ratings or follow-up requests.	Citizen, System	Medium
FR-09	The system shall provide dashboards and reports tailored to each role.	All roles	High
FR-10	The system shall send a daily overdue-complaint alert at 09:00 Indian Standard Time.	System	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Security	Access is limited to authenticated users through Zoho Creator profiles; citizen e-mail and phone numbers are flagged as personal data.
NFR-02	Usability	Picklists and lookups replace free text wherever possible; phone and tablet layouts are defined.
NFR-03	Availability	The application is hosted on Zoho's cloud and accessible wherever an internet connection is available.
NFR-04	Maintainability	Status logic is kept in named form workflows; field link names follow a consistent pattern.
NFR-05	Auditability	System fields record the creator and time of each record; workflow runs appear in the application logs.
NFR-06	Time zone	Dates and times use Asia/Kolkata and the 24-hour clock.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (India data centre, creator.zoho.in)
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
Mobile	Zoho Creator mobile app (Android / iOS)
Other Zoho service	Zoho Creator sendmail task and AI sentiment-analysis field
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

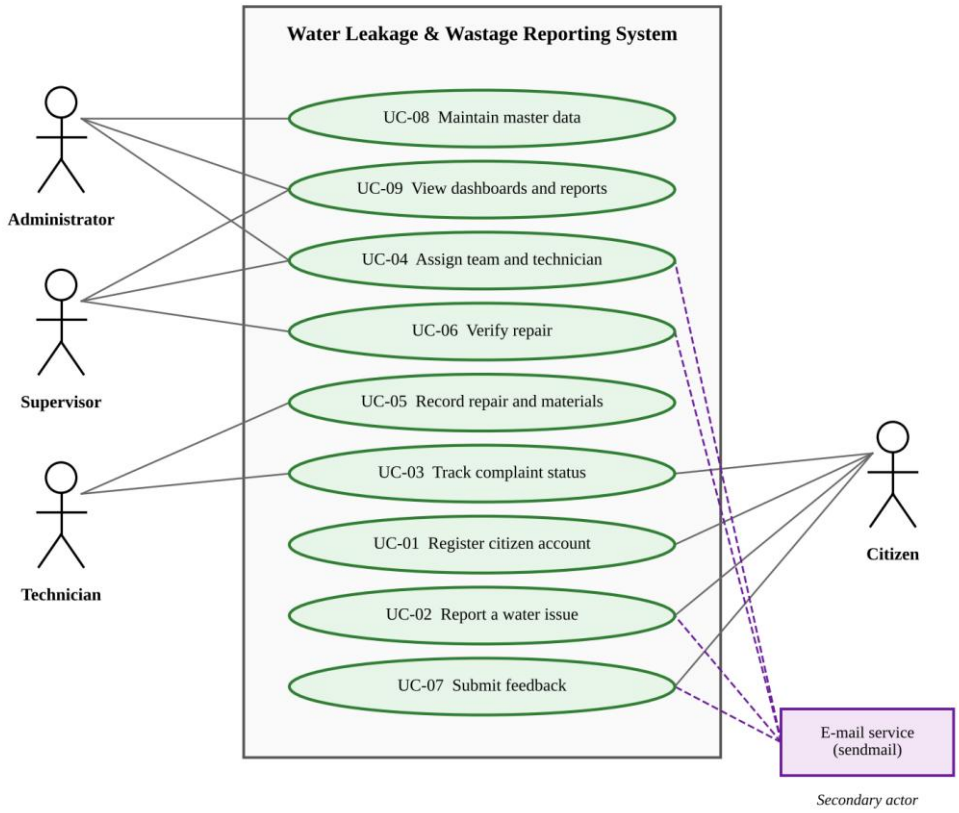


Figure 3.1: Use Case Diagram of the Water Leakage & Wastage Reporting Management System

Table 3.5: Use Case Description – Report a Water Issue

Item	Description
Use Case ID	UC-01
Use Case Name	Report a water issue
Actor(s)	Citizen (primary), System
Pre-condition	The citizen is registered with an active account and the Citizen profile is assigned.
Main Flow	1. The citizen opens Water Issues > Water Issue Report and enters the title, issue type, district, locality, street address, landmark, description and photos. 2. The citizen selects severity and priority. 3. On submission the status is set to submitted and the report date and time are filled automatically. 4. The confirmation workflow sends an e-mail to the citizen and the administrator. 5. The complaint appears in the citizen's My Complaints report and in the administrator's dashboards.
Alternate Flow	2a. A mandatory field is empty: the form rejects the submission and highlights the field.
Post-condition	A water issue record with status submitted exists and is visible to supervisors.
Related FR	FR-02, FR-03, FR-09

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

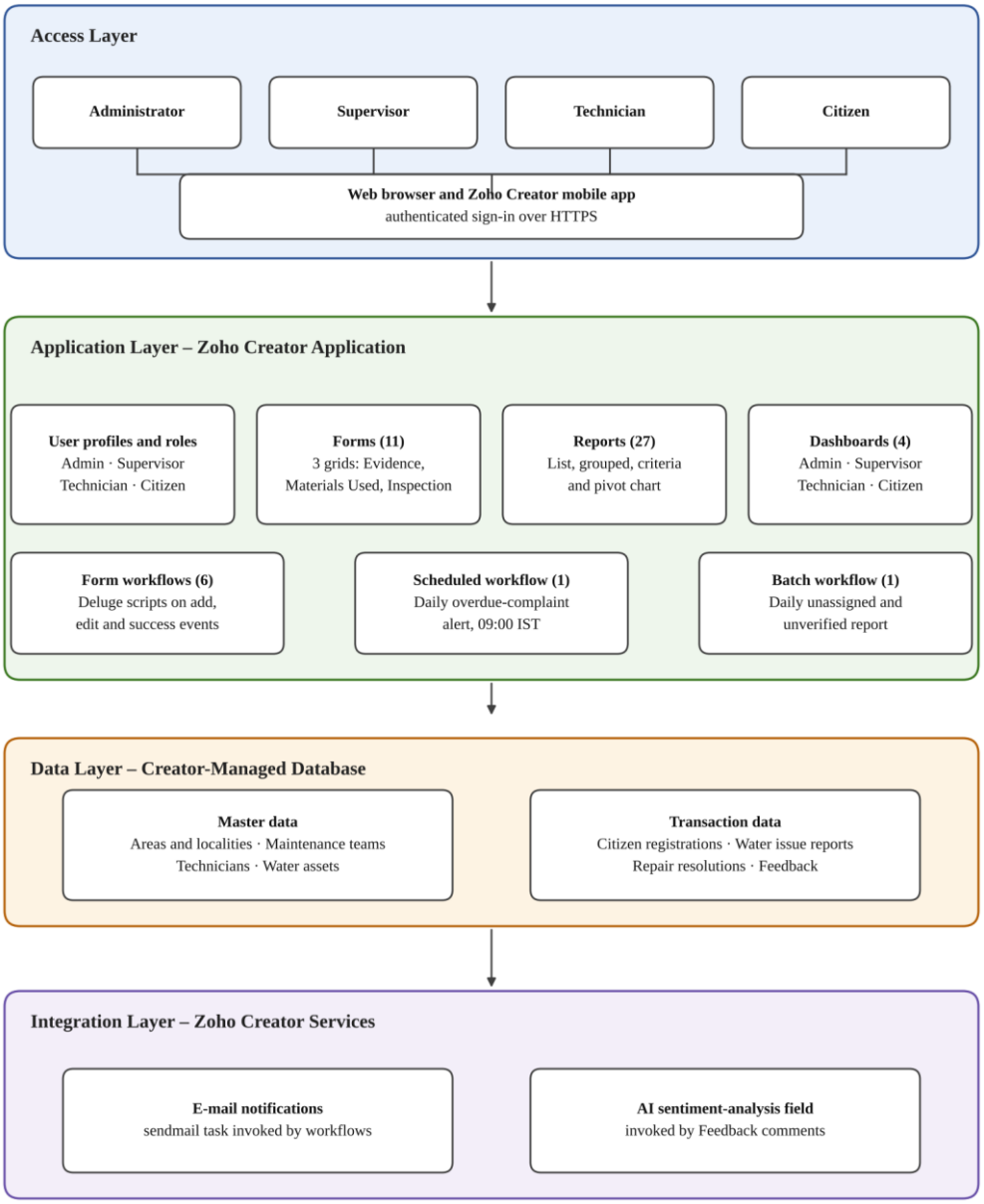


Figure 4.1: System Architecture of the Water Leakage & Wastage Reporting Management System

The application is organised in four layers. The access layer consists of the users and their channels: administrators, supervisors and technicians work on the Zoho Creator web interface or the mobile app, and citizens use the web or the mobile app with a Citizen profile. The application layer is the Zoho Creator application itself: eleven forms, 27 reports, four dashboard pages, six form workflows, one scheduled workflow

and one batch workflow and the Deluge scripts. Profiles and roles at this layer decide what each user can see and change.

The data layer is Zoho Creator's managed database, in which forms are connected by lookup fields and grids. The integration layer uses Zoho Creator's own services: the sendmail task for e-mail notifications and the AI sentiment-analysis field on the feedback form.

4.2 Data Model

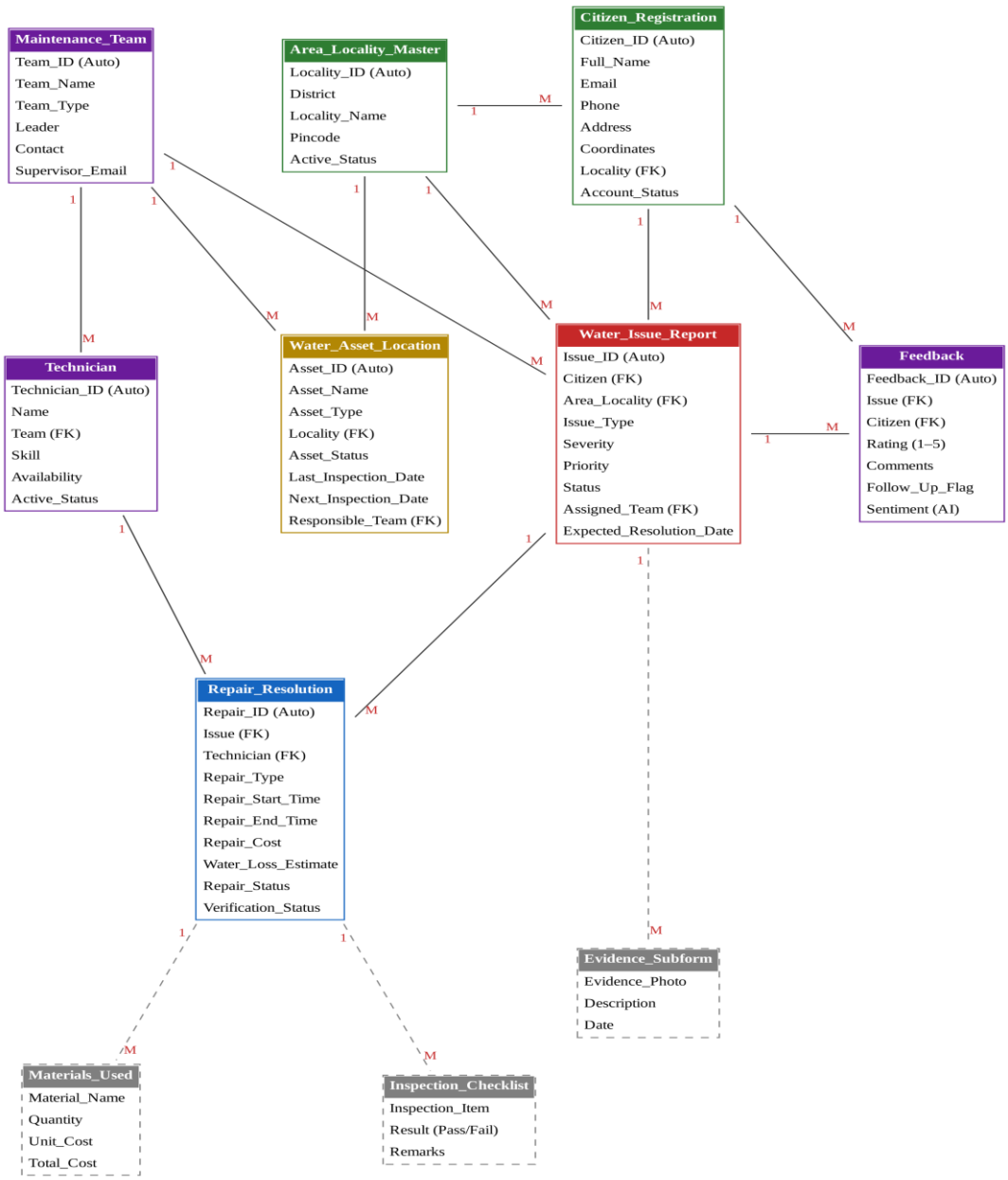


Figure 4.2: Entity–Relationship Diagram of the Water Leakage & Wastage Reporting Management System

Table 4.1: Entities and Relationships

Parent Form	Child / Related Form	Relationship	Implemented Using
Area_Locality_Master	Citizen_Registration	One-to-Many	Lookup (Area_Locality_Master) in Citizen_Registration
Area_Locality_Master	Water_Issue_Report	One-to-Many	Lookup (Area_Locality) in Water_Issue_Report
Area_Locality_Master	Water_Asset_Location	One-to-Many	Lookup (Area_Locality) in Water_Asset_Location
Citizen_Registration	Water_Issue_Report	One-to-Many	Lookup (Citizen) in Water_Issue_Report
Citizen_Registration	Feedback	One-to-Many	Lookup (Citizen) in Feedback
Water_Issue_Report	Repair_Resolution	One-to-Many	Lookup (Issue) in Repair_Resolution
Water_Issue_Report	Feedback	One-to-Many	Lookup (Issue) in Feedback
Water_Issue_Report	Evidence_Subform	One-to-Many	Grid (Evidence_Subform) in Water_Issue_Report
Maintenance_Team	Technician	One-to-Many	Lookup (Team) in Technician
Maintenance_Team	Water_Asset_Location	One-to-Many	Lookup (Responsible_Team) in Water_Asset_Location
Maintenance_Team	Water_Issue_Report	One-to-Many	Lookup (Assigned_Team) in Water_Issue_Report
Technician	Repair_Resolution	One-to-Many	Lookup (Technician) in Repair_Resolution
Repair_Resolution	Materials_Used	One-to-Many	Grid (Materials_Used) in Repair_Resolution
Repair_Resolution	Inspection_Checklist	One-to-Many	Grid (Inspection_Checklist) in Repair_Resolution

The Water Issue Report is the centre of the model. Every complaint refers to a citizen and an area through lookups, can be assigned to one team and one technician, and collects its photos in the Evidence grid. Each repair record refers to one complaint and one technician, and holds its materials and checklist as grids. Feedback refers to both the citizen and the complaint it evaluates.

4.3 Form Design

Table 4.2: Field Specification – Water Issue Report Form (key fields)

Field Label	Link Name	Field Type	Mandatory	Validation / Remarks
Issue ID	Issue_ID	Auto number	—	Generated on submission
Citizen	Citizen	Lookup	No	Shows citizen full name
Issue Title	Issue_Title	Single line	Yes	Short description of the problem
Issue Type	Issue_Type	Drop down	No	Water Pipe Leakage, Pipeline Burst, Public Tap Leakage, Overflowing Water Tank, Water Wastage, Broken Water Valve, Blocked Water Pipeline, Low/No Water Supply, Contaminated Water, Damaged Water Meter, Other
District	District	Drop down	No	38 districts of Tamil Nadu
Area / Locality	Area_Locality	Lookup	No	Shows locality name
Street Address / Pincode / Landmark	Street_Address, Pincode, Landmark	Text	No	Free text
Issue Photo	Issue_Photo	Image	No	Up to 10 files; camera and gallery
Report Date / Time	Report_Date, Report_Time	Date, time	No	Filled with current date and time
Severity	Severity	Drop down	No	Minor, Moderate, Major, Critical
Priority	Priority	Drop down	No	Low, Medium, High, Critical
Status	Status	Drop down	No	Default submitted; also Under Review, Approved, Assigned, In Progress, On Hold, Resolved, Rejected, Closed
Assigned Team / Technician	Assigned_Team, Technician	Lookup	No	Set by supervisor
Expected Resolution Date	Expected_Resolution_Date	Date	No	Used by overdue alert
Resolution Notes	Resolution_Notes	Multi line	No	Copied from verification remarks
Evidence	Evidence_Subform	Grid	No	Photos and descriptions

Table 4.3: Field Specification – Repair Resolution Form (key fields)

Field Label	Link Name	Field Type	Remarks
Repair ID	Repair_ID	Auto number	Generated on submission
Issue	Issue	Lookup to Water Issue Report	Complaint being repaired
Technician	Technician	Lookup to Technician	Person performing the repair
Repair Start / End Time	Repair_Start_Time, Repair_End_Time	Date-time	hh:mm:ss
Repair Type	Repair_Type	Drop down	Pipeline Repair, Leakage Repair, Valve Replacement, Pipe Replacement, Tank Repair, Pump Repair, Meter Repair, Cleaning, Other
Action Taken	Action_Taken	Multi line	Description of work
Repair Cost	Repair_Cost	Currency (INR)	Indian digit grouping
Water Loss Estimate	Water_Loss_Estimate	Decimal	Estimated water lost
Repair Photo	Repair_Photo	Image	Up to 10 files
Inspection Result	Inspection_Result	Drop down	Passed, Failed, Requires Further Work
Repair Status	Repair_Status	Drop down	Not Started, In Progress, On Hold, Completed (drives complaint status)
Verified By / Verification Status	Verified_By, Verification_Status	Text, drop down	Pending, Approved, Rework Required (drives complaint status)
Materials Used	Materials_Used	Grid	Material name, quantity, unit cost, total cost
Inspection Checklist	Inspection_Checklist	Grid	Inspection item, Pass/Fail, remarks

Table 4.4: Summary of Other Forms

Form	Principal Fields	Notes
Area_Locality_Master	Locality ID, district, locality name, pincode, active status	Master list used by lookups
Citizen_Registration	Citizen ID, full name (prefix, first, last), e-mail, phone, address with coordinates, district, locality, registration date, account status	Name, e-mail, phone and address flagged as personal data

Form	Principal Fields	Notes
Feedback	Feedback ID, issue, citizen, rating (1–5), comments, feedback date, follow-up flag, sentiment	Sentiment is an AI field based on Comments
Maintenance_Team	Team ID, name, department, team type, leader, contact, supervisor e-mail, district, locality, size, availability, status	Six team types
Technician	Technician ID, name, e-mail, phone, team, skill, district, locality, availability, active status, joining date	Name flagged as personal data
Water_Asset_Location	Asset ID, name, type, district, locality, street, installation date, asset status, last and next inspection dates, responsible team	Eight asset types
Materials_Used	Material name, quantity, unit cost, total cost (formula: quantity × unit cost)	Used as a grid in Repair Resolution
Inspection_Checklist	Inspection item, result (Pass/Fail), remarks	Used as a grid in Repair Resolution
Evidence_Subform	Evidence photo, description, date	Used as a grid in Water Issue Report

4.4 Process and Workflow Design

The central process is the life cycle of a water complaint. A complaint is created with the status submitted. When a supervisor assigns a team and technician, the assignment workflow sets the status to Assigned and e-mails the technician. The repair workflow maps the repair status onto the complaint: In Progress and On Hold are copied to the complaint, and Completed moves the complaint to Under Review and e-mails the supervisor of the assigned team (or the administrator if the team has no supervisor e-mail) with the repair details. A further form workflow, Supervisor Verification Email, e-mails the team supervisor whenever a repair record's status is updated, which duplicates the completion e-mail (see BUG-09 in Table 6.2). A batch workflow, Daily Assignment and Verification Report, runs once a day and e-mails the administrator a list of complaints without a technician and repairs still awaiting verification. The verification workflow then either resolves the complaint (Approved, with the remarks copied to Resolution Notes and the citizen informed) or returns it to In Progress (Rework Required, with the technician informed). Feedback forms are checked on submission and an alert is sent to the administrator when the rating is 1 or 2 or follow-up is requested.

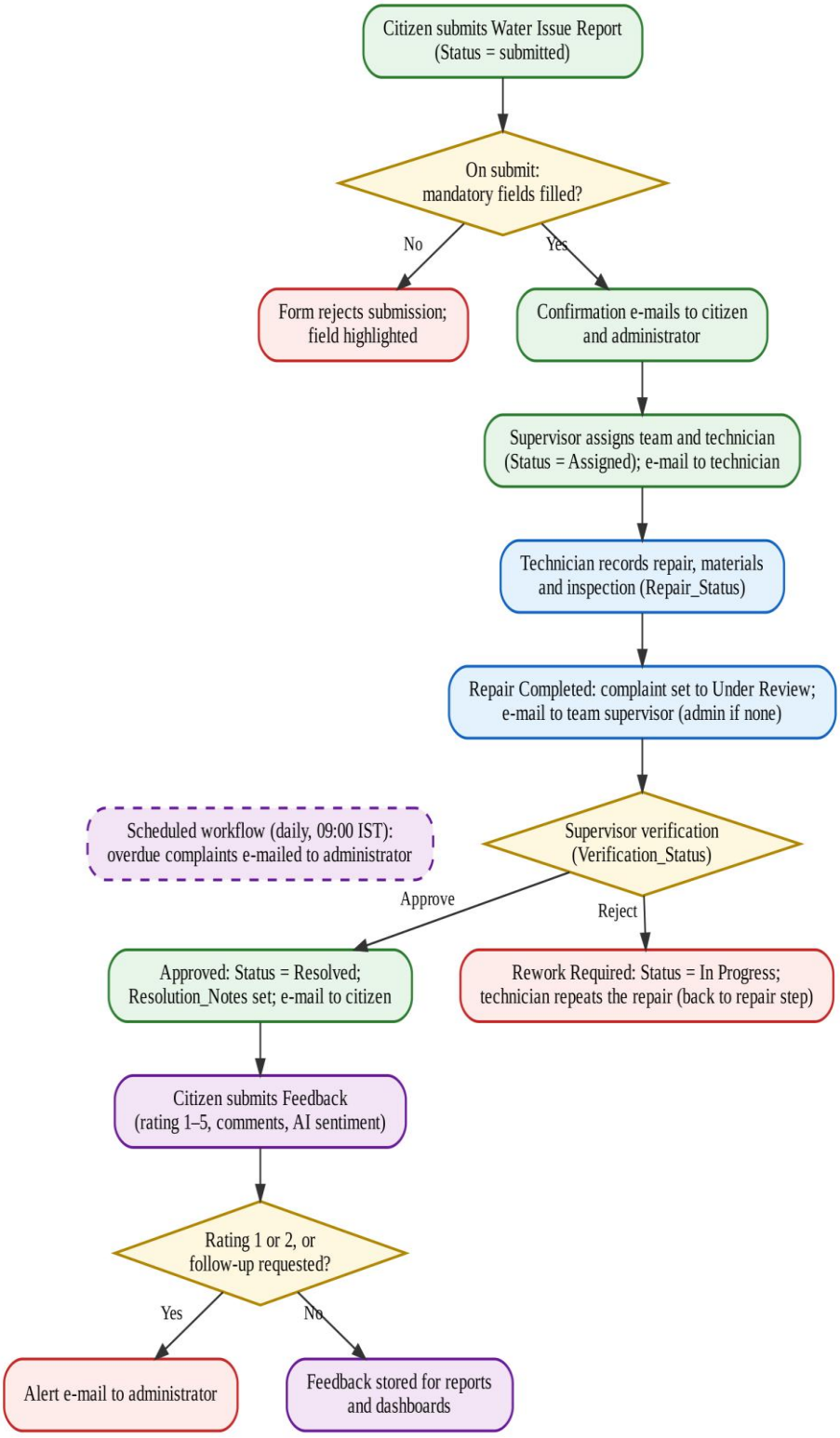


Figure 4.3: Complaint Status Life Cycle and Workflow Triggers

4.5 User Interface and Navigation Design

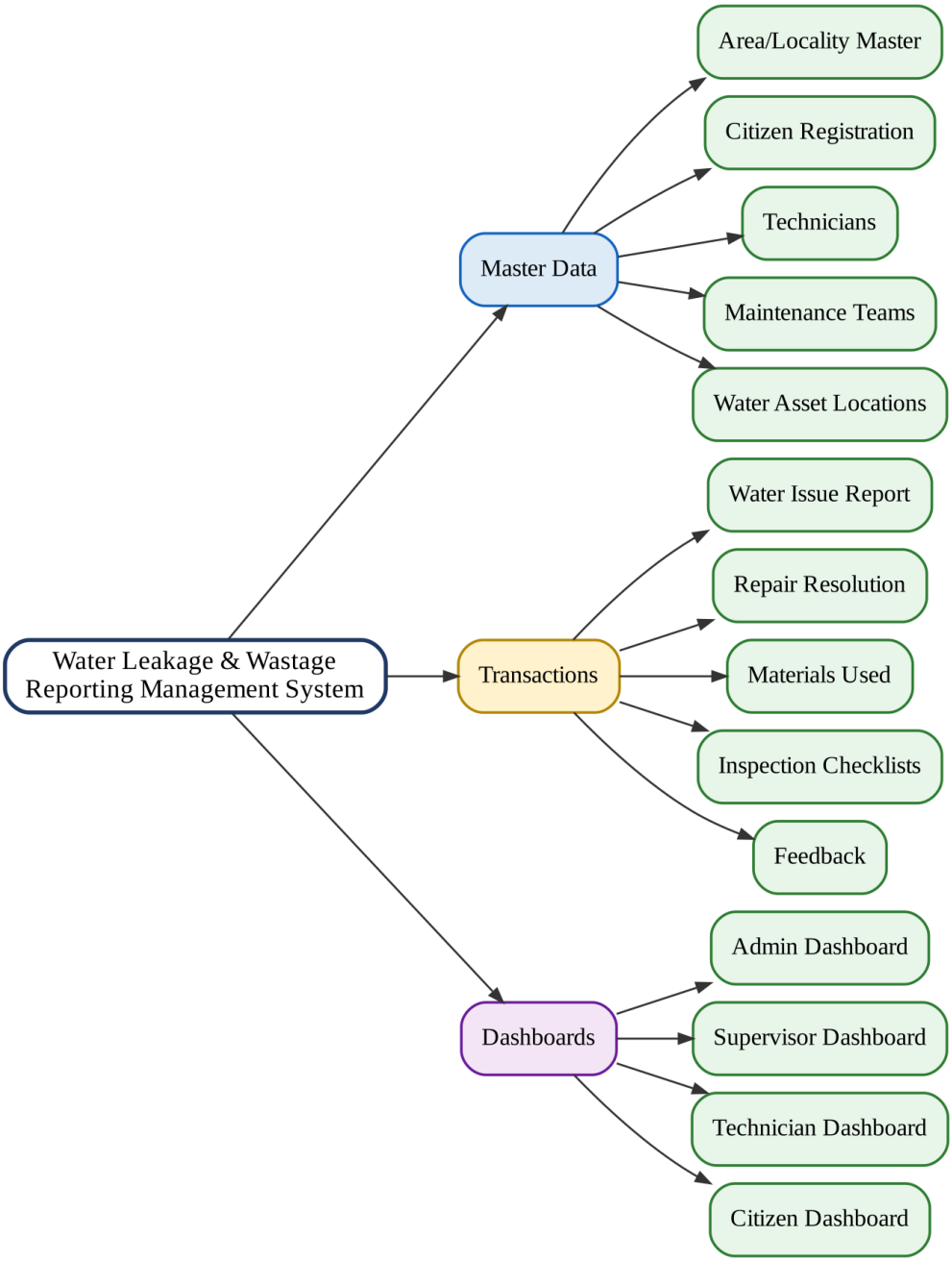


Figure 4.4: Navigation Structure of the Application

The navigation menu has one section for each subject area: Area/Locality Master, Citizen Registration, Water Issues (report, all issues, pending and high-priority views, district and locality views), Repair Resolution, Technicians, Maintenance Teams, Water Assets, Materials Used, Inspection Checklists, Feedback, and the four dashboard pages. Zoho Creator hides components for which a user has no permission, so the same structure presents a smaller menu to restricted users.

4.6 Access Control Design

Access is designed around four roles arranged in a hierarchy: Admin, with Supervisor and Technician below it, and Citizen beside them. Each role is mapped to a Zoho Creator permission profile. Admin_Permissions gives full create, view, modify, import and export rights on the master and transaction forms. Supervisor_Permissions allows review and assignment, while keeping the complaint description and location read-only. Technician_Permissions lets field staff see assigned complaints and update only repair fields. Citizen_Permissions allows citizens to create complaints and feedback and to see only their own records, using report criteria on the logged-in e-mail address. Table 4.5 summarises the intended access.

Table 4.5: Role–Permission Matrix (Intended Design)

Component	Administrator	Supervisor	Technician	Citizen
Master forms (Area, Maintenance Team, Technician)	Full (A E V D)	View, edit selected fields	View own details	Own details only
Citizen Registration	Full	Create, view, modify	View only	Own record
Water Issue Report	Full	Create, assign, set status and priority	View assigned (read-only fields)	Create; view own
Repair Resolution	Full	Verify; view	Create and edit repair fields	No access
Water Asset Location	Full	Create, view, modify	View only	View only
Feedback	Full	View	No access	Create; view own
Dashboards	Admin Dashboard	Supervisor Dashboard	Technician Dashboard	Citizen Dashboard

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application “Water Leakage & Wastage Reporting Management System” was created on Zoho Creator in the India data centre. It uses the date format dd-MMM-yyyy, the 24-hour time format and the Asia/Kolkata time zone. Its icon is the text “WLR”. Components are grouped into navigation sections. A configuration export (.ds file, version 1.0, generated on 27 October 2026) was taken and is the source for the figures and code quoted in this report.

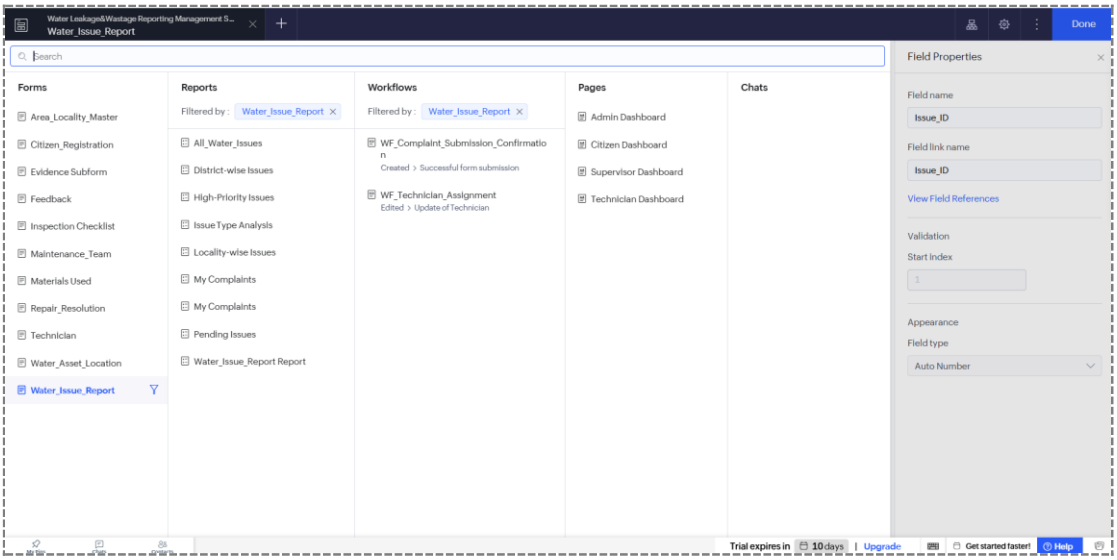


Figure 5.1: Application Home – Navigation Sidebar with Sections Expanded

5.2 Forms and Data Capture

Eleven forms were built with the Zoho Creator form builder (Chapter 4). They use text, number, decimal, date, date-time, e-mail, phone, address, image, multi-line, radio button, checkbox, drop-down, lookup, currency, grid and AI fields. Mandatory fields are marked, drop-downs restrict values to controlled lists, and lookups to the master forms guarantee that every complaint refers to an existing citizen, area and team. Cost fields use Indian rupee currency with Indian digit grouping. Each form shows a success message after submission, such as “Water Issue Report Added Successfully” or “Feedback Added Successfully”.

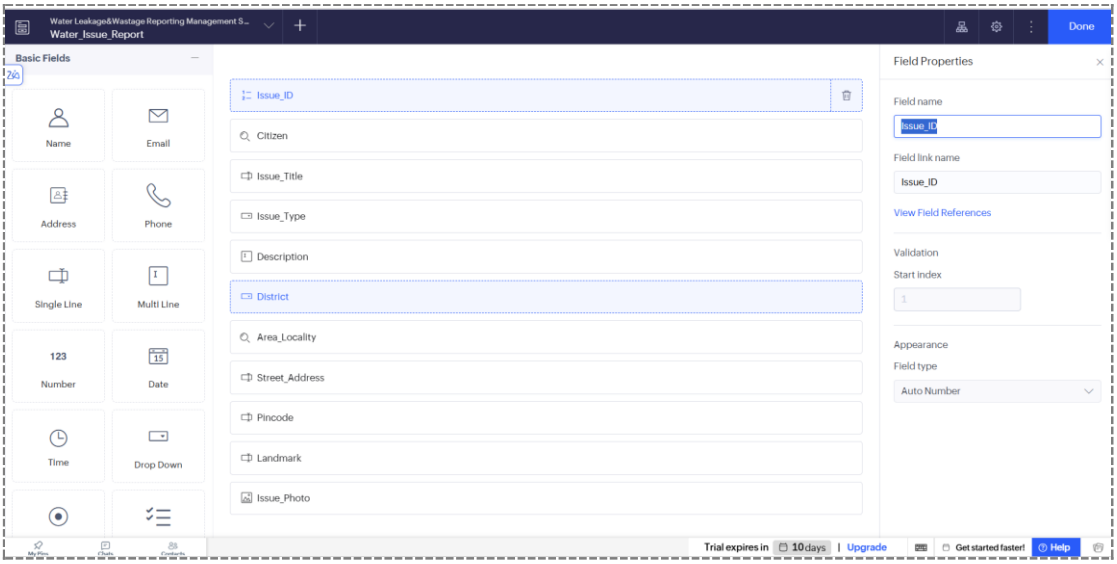


Figure 5.2: Water Issue Report Form in the Form Builder

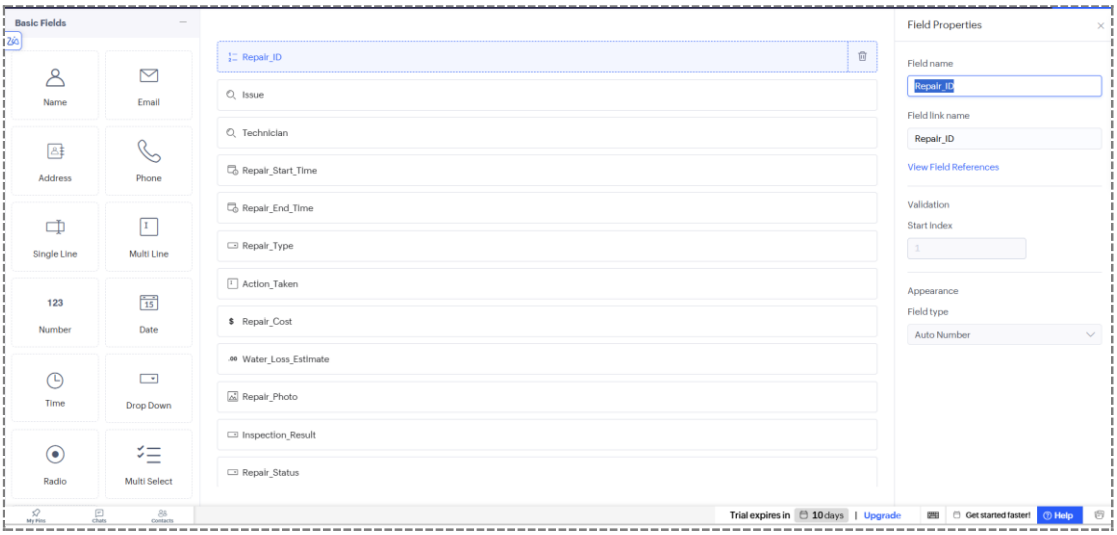


Figure 5.3: Repair Resolution Form with Materials and Inspection Grids

5.3 Reports

Twenty-seven reports were configured: twenty-six list reports and one pivot chart, the Issue Type Split, which summarises water issues by issue type. Most are list reports with columns chosen for each audience. Several use grouping (district-wise, locality-wise, issue-type, asset-type, sentiment and technician), two apply criteria so that users see only their own data, and one shows the technician's own details. Table 5.1 groups them by purpose.

Table 5.1: Reports Implemented

Group	Reports	Purpose
Master data	Area/Locality Master Report; Citizen Registration Report; Maintenance Team Report; All Technicians; Technician Details (criteria: own e-mail)	Maintain and review masters
Water issues	All Water Issues; Water Issue Report; Pending Issues (excludes resolved, closed, rejected); High-Priority Issues; District-wise Issues; Locality-wise Issues; Issue Type Analysis; Issue Type Split (pivot chart); My Complaints; My Complaints (citizen view, criteria: own login)	Track and analyse complaints
Repairs	Repair Resolution Report; Technician Performance (completed repairs, grouped by technician); Resolution Time (sorted by end time)	Measure repair work
Assets and materials	Water Asset Location Report; Water Asset Register (grouped by asset type); Materials Used Report; All Inspection Checklists; Evidence Subform Report	Asset and materials records
Feedback	All Feedback; Citizen Feedback (grouped by sentiment); My Feedback (criteria: own e-mail); Own Citizen Registration	Service quality



Figure 5.4: : Issue Type Analysis Report with Filters

5.4 Pages and Dashboards

Four dashboard pages were built with the Zoho Creator page builder, one for each role. Each page binds KPI panels, charts and embedded reports directly to the forms, so the figures reflect current data. Table 5.2 lists the components.

Table 5.2: Components of the Dashboards

Dashboard	KPI Panels	Charts	Embedded Reports / Forms
Admin Dashboard	Total issues; pending issues; high-priority issues; resolved complaints; overdue complaints; repairs awaiting verification	Complaints by district (column); complaints by status (pie); water issues by type (column)	High-priority issues; technician performance; all water issues; pending issues; high-priority issues; water asset register; citizen feedback
Supervisor Dashboard	Complaints to review; high-priority complaints; active repairs; repairs awaiting verification	Complaints by locality (column); repairs by status (pie)	Pending issues; high-priority issues; repair resolution report; locality-wise issues; technician performance; resolution time
Technician Dashboard	Assigned issues; repairs in progress; completed repairs	None	My Complaints; repair resolution form and report; water asset register
Citizen Dashboard	Pending complaints; resolved complaints; total complaints	Water issues by type (pie)	Complaint form; feedback form; My Complaints report

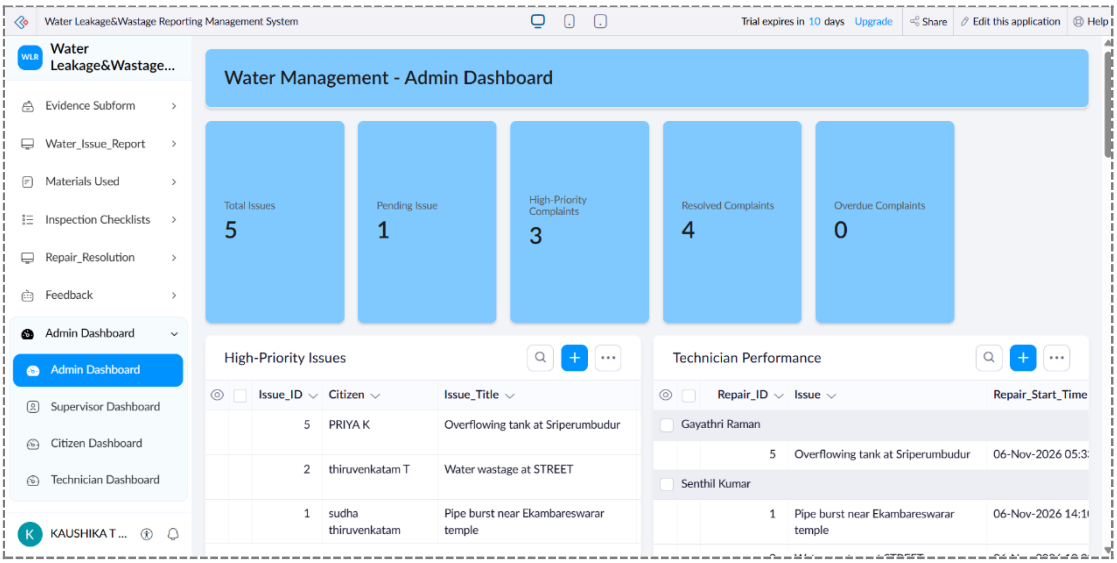


Figure 5.5: Admin Dashboard

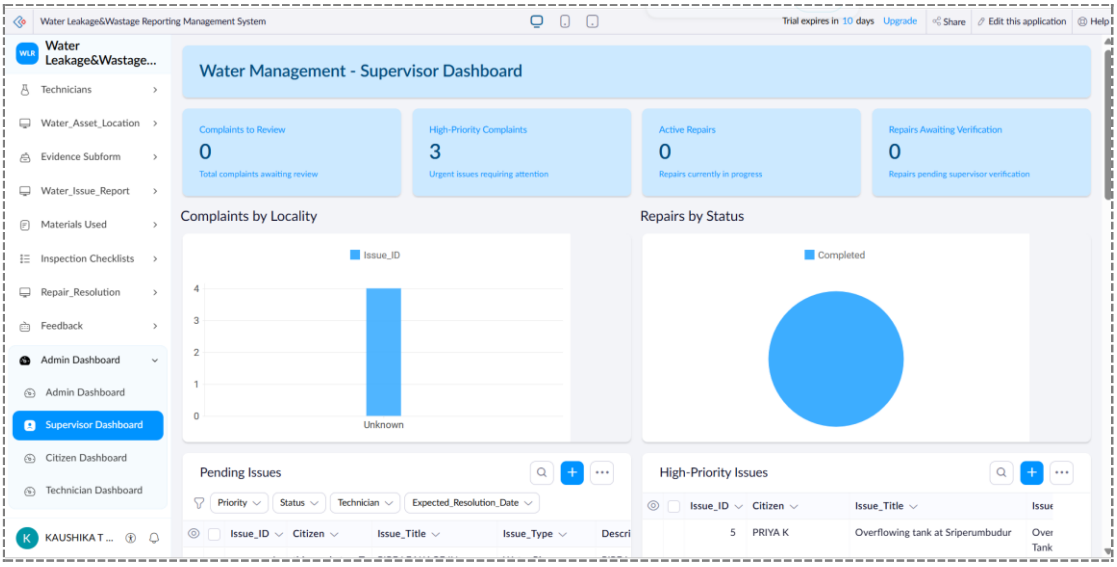


Figure 5.6: Supervisor Dashboard

5.5 Deluge Scripting

Deluge, Zoho's scripting language, is used in the form workflows, the scheduled workflow and the batch workflow. The scripts are summarised in Table 5.3 and reproduced in Appendix C.

Table 5.3: Deluge Scripts Implemented

ID	Workflow	Type	Description
DS-01	WF_Complaint_Submission_Confirmation	Form workflow (on success, Water Issue Report)	Sends a confirmation e-mail to the administrator and the citizen with the issue ID, title and status
DS-02	WF_Technician_Assignment	Form workflow (on edit, update of Technician)	Sets status to Assigned when a technician is chosen and e-mails the technician
DS-03	WF_Repair_Progress	Form workflow (on edit, update of Repair Status, Repair Resolution)	Maps In Progress and On Hold onto the complaint; on Completed sets Under Review and e-mails the team supervisor with the repair details
DS-04	WF_Repair_Verification	Form workflow (on edit, update of Verification Status)	Approved: Resolved, copies remarks and informs citizen. Rework Required: In Progress and informs technician

ID	Workflow	Type	Description
DS-05	WF_Feedback_Follow_Up	Form workflow (on success, Feedback)	E-mails the administrator when follow-up is requested or the rating is 1 or 2
DS-06	WF_Daily_Overdue_Alert	Scheduled workflow, daily at 09:00 Asia/Kolkata from 20 October 2026	Sends the locality-wise issues report as PDF and one e-mail per overdue complaint
DS-07	Supervisor_Verification_Email	Form workflow (on edit, update of Repair Status, Repair Resolution)	Sends “Repair Awaiting Verification” to the team supervisor on every update of Repair Status (see BUG-09)
DS-08	Daily_Assignment_and_Verification_Report	Batch workflow, daily (Water Issue Report and Repair Resolution)	Sends the administrator a list of unassigned complaints and repairs with Verification Status Pending

Code 5.1: WF_Repair_Progress (form workflow, on edit of Repair_Status, excerpt)

```
if(input.Repair_Status == "Completed")
{
complaint.Status="Under Review";
supervisorEmail = zoho.adminuserid;
if(complaint.Assigned_Team != null)
{
team = Maintenance_Team[ID == complaint.Assigned_Team];
if(team.count() > 0 && team.Supervisor_Email != null)
{
supervisorEmail = team.Supervisor_Email;
}
}
technicianName = "Not assigned";
if(input.Technician != null)
{
technicianName = input.Technician.Technician_Name.first_name;
}
sendmail
[
from :zoho.adminuserid
to :supervisorEmail
subject :"Repair Awaiting Verification"
message :"A repair has been marked Completed and needs your
verification.\n\nRepair ID: " + input.Repair_ID + "\nIssue ID: " +
complaint.Issue_ID + "\nIssue: " + complaint.Issue_Title +
"\nTechnician: " + technicianName + "\nAction Taken: " +
ifnull(input.Action_Taken,"") + "\nRepair Cost: " +
ifnull(input.Repair_Cost,"") + "\nWater Loss Estimate: " +
ifnull(input.Water_Loss_Estimate,"") + "\n\nOpen Repair_Resolution to
set Verification Status to Approved or Rework Required."
]
```

```
}
```

Code 5.2: WF_Technician_Assignment (form workflow, on update of Technician, excerpt)

```
if(input.Technician != null)
{
 input.Status = "Assigned";
 sendmail [ from: zoho.adminuserid  to: input.Technician.Email
subject: "New Water Repair Assignment"  message: "..." ]
}
```

5.6 Workflows and Automation

Table 5.4: Workflows Implemented

Workflow	Form	Trigger	Action
WF_Complaint_Submission_Confirmation	Water Issue Report	On success of add	Confirmation e-mails (DS-01)
WF_Technician_Assignment	Water Issue Report	On edit, update of Technician	Status Assigned; technician e-mail (DS-02)
WF_Repair_Progress	Repair Resolution	On edit, update of Repair Status	Complaint status update; team supervisor e-mail on completion (DS-03)
WF_Repair_Verification	Repair Resolution	On edit, update of Verification Status	Resolved or In Progress; citizen or technician e-mail (DS-04)
WF_Feedback_Follow_Up	Feedback	On success of add	Administrator alert for follow-up or low rating (DS-05)
WF_Daily_Overdue_Alert	Scheduled	Daily at 09:00 Asia/Kolkata	PDF report and overdue e-mails (DS-06)
Supervisor_Verification_Email	Repair Resolution	On edit, update of Repair Status	Team supervisor e-mail on every status update (DS-07)
Daily_Assignment_and_Verification_Report	Water Issue Report, Repair Resolution	Batch, daily	Administrator digest of unassigned complaints and unverified repairs (DS-08)

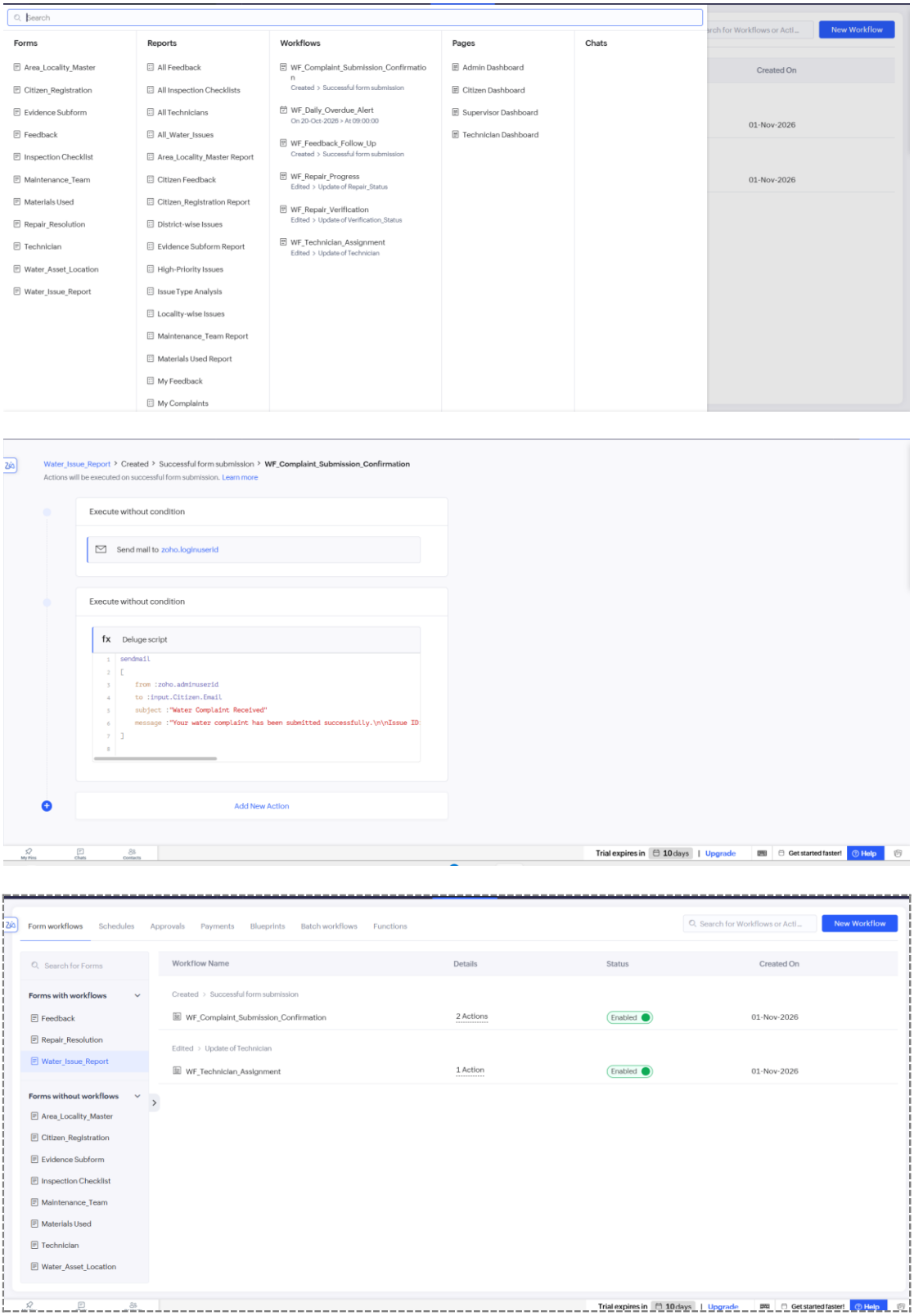


Figure 5.7: Workflow List of the Application

Form workflows are used for status changes so that the complaint status is updated in the same submission as the repair or verification. Two workflows run once a day: the

scheduled overdue alert (WF_Daily_Overdue_Alert) and the batch Daily Assignment and Verification Report.

5.7 Integrations and Notifications

Table 5.5: Integrations and Notifications Implemented

Feature	Mechanism	Used By	Purpose
E-mail notifications	Deluge sendmail task from the workflows	DS-01 to DS-08	Inform citizens, technicians and supervisors at each step
Overdue report	Locality-wise report attached as PDF to a scheduled e-mail	DS-06	Give the administrator a daily list of locality-level issues
Sentiment analysis	AI field of type SentimentAnalysis on the Comments field of Feedback	Feedback form and Citizen Feedback report	Classify feedback comments by sentiment

No external connection, REST call or CRM integration is used in this version. Notifications rely entirely on the platform's own mail task.

5.8 User Management and Role-Based Access

Table 5.6: Profiles and Roles of the Application

Profile / Role	Purpose	Key Permissions
Administrator	Owner with all permissions	All components
Admin_Permissions	Admin role	Create, view, modify, import, export and report on master and transaction forms; Admin Dashboard tab
Supervisor_Permissions	Supervisor role	Create, view and modify citizen, asset and repair records; assign and set status on complaints; Supervisor Dashboard tab
Technician_Permissions	Technician role	View assigned complaints; create and edit repair fields and grids; Technician Dashboard tab
Citizen_Permissions	Citizen role	Create and view own complaints and feedback; edit own registration; Citizen Dashboard tab
Read / Write / Developer / Customer	Platform default profiles	Available for additional users; the Customer profile has add and view rights

Roles form a hierarchy: Admin is the top role, Supervisor and Technician are its children, and Citizen is a separate branch under Admin. Users are invited and assigned

a profile, and the profile determines whether each component can be added, edited or viewed. Citizen and technician records are flagged as personal data.

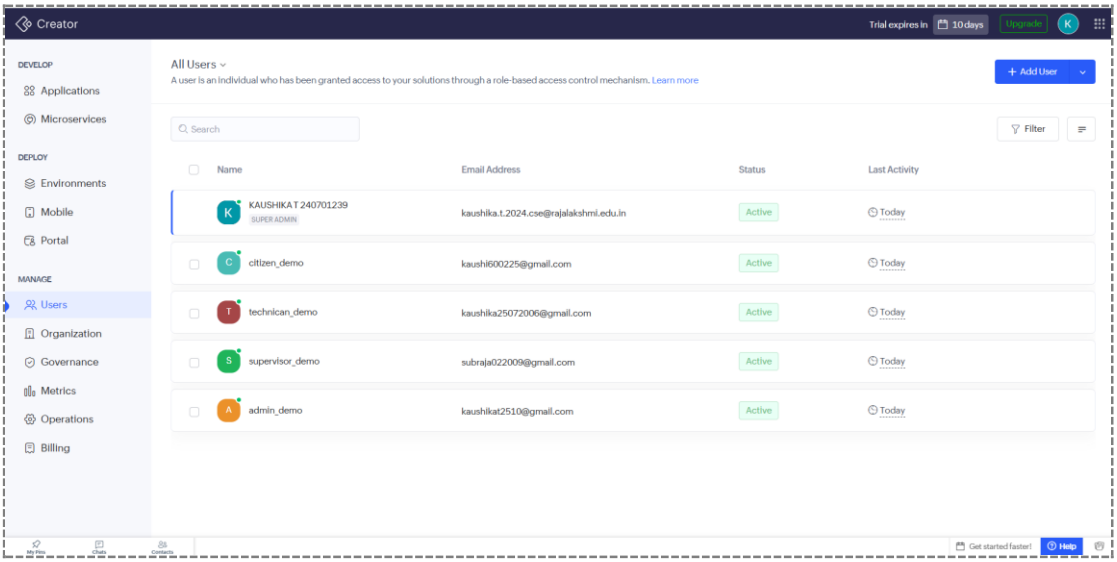


Figure 5.8: Profiles and Roles of the Application

5.9 AI and Analytics Features

Two features provide intelligence. First, the Sentiment field on the Feedback form is an AI field based on the Comments text. It classifies each comment and allows citizen feedback to be grouped by sentiment in the Citizen Feedback report. Second, the dashboards and reports provide analytics through charts, KPI panels and grouped reports. Figure 5.9 shows the feedback form.

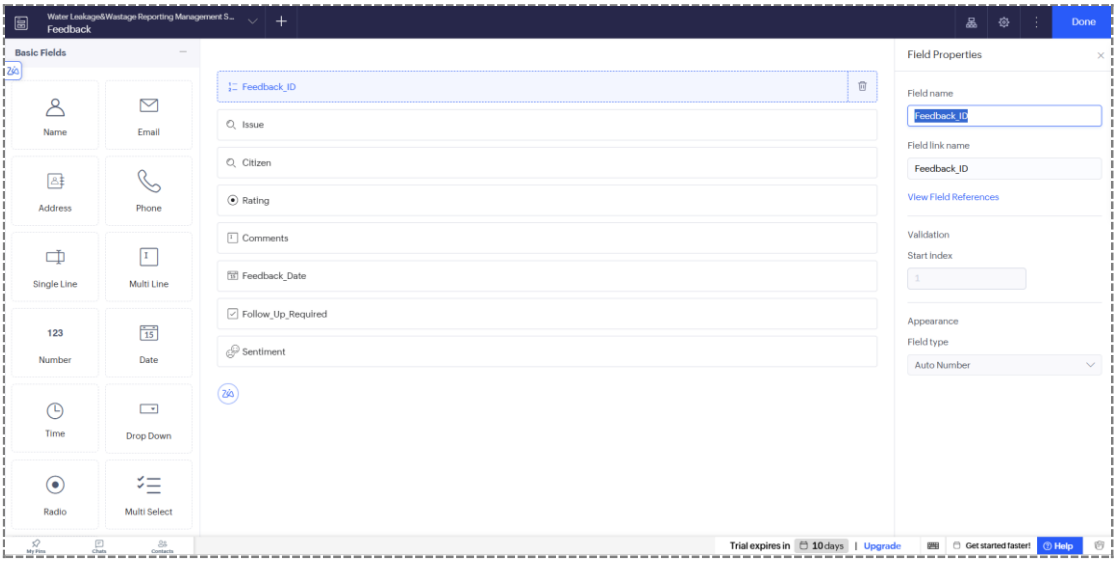


Figure 5.9: Feedback Form with Sentiment Field

5.10 Pro-Code Extension

Pro-code extension was limited to Deluge scripting inside the form workflows and the scheduled workflow (Section 5.5, Appendix C). No external REST API, custom widget, separate programming stack or Zoho CRM integration was built in this version, because the requirements were met by the platform's own mail task, reports and dashboards. The section is retained to show this decision.

5.11 Mobile and Portal Access

The application defines separate layouts for the web, phone and tablet. The phone layout uses a two-column arrangement with automatic label placement, and the tablet layout uses a sliding pane. Citizens can therefore report problems from a phone on site, and technicians can record repairs in the field. Figure 5.10 shows the application on the mobile app.

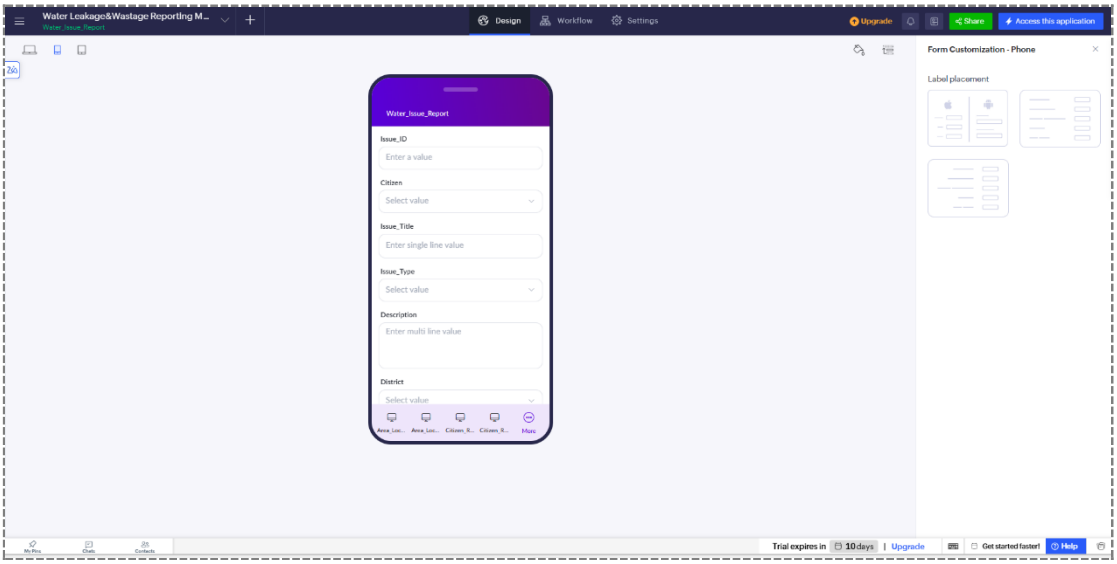


Figure 5.10: Application on the Zoho Creator Mobile App

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing is organised at four levels. Unit testing checks each form and workflow in isolation: each form is submitted in live mode and each workflow is confirmed by the record it changes or the e-mail it sends. Integration testing covers the paths that cross components: a complaint must move through assignment, repair, verification and resolution, and each step must update the dashboards and reports. Access testing logs in with users of each profile and checks what each can see and change. User acceptance testing is described in Section 6.5. Every test case is traced to a functional requirement in Table 6.1.

The Status column of Table 6.1 is intentionally left for the tester to complete (Pass or Fail) while the test is executed in live mode, and the actual values should be attached as screenshots.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Status (Pass / Fail)
TC-01	FR-01	Register a citizen with all mandatory fields	Enter name, e-mail, phone, address, district, locality; submit	Record saved with Citizen ID and Active status	pass
TC-02	FR-01	Locality lookup filtered by district	Open Citizen Registration; select a locality	Only localities of the selected area offered, sorted ascending	pass
TC-03	FR-02	Report a water issue	Enter title, type, severity, priority, locality, photo; submit	Status submitted; report date and time filled; success message shown	pass
TC-04	FR-02	Mandatory validation on Water Issue Report	Leave Issue Title empty; submit	Form rejects submission and highlights the field	pass
TC-05	FR-02	Attach evidence photos	Add up to 10 files in the Evidence grid	All files saved and visible in the record	pass
TC-06	FR-03	Confirmation e-mails on submission	Submit TC-03 record	E-mail received by citizen and administrator with issue ID and status	pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Status (Pass / Fail)
TC-07	FR-04	Assign technician	Edit complaint; choose team and technician	Status Assigned; technician receives assignment e-mail	pass
TC-08	FR-06	Repair in progress	Create repair with status In Progress	Complaint status In Progress	pass
TC-09	FR-06	Repair on hold	Set repair status On Hold	Complaint status On Hold	pass
TC-10	FR-06	Repair completed	Set repair status Completed	Complaint status Under Review; supervisor receives verification e-mail	pass
TC-11	FR-06	Verification approved	Set verification status Approved; enter remarks	Complaint Resolved; remarks in Resolution Notes; citizen e-mailed	pass
TC-12	FR-06	Verification rework	Set verification status Rework Required	Complaint In Progress; technician e-mailed	pass
TC-13	FR-05	Materials total cost	Add material with quantity 4 and unit cost 250	Total cost 1,000 calculated	pass
TC-14	FR-05	Inspection checklist grid	Add two checklist rows with Pass and Fail	Both rows saved against the repair	pass
TC-15	FR-08	Feedback with low rating	Submit feedback with rating 1	Administrator receives follow-up e-mail	pass
TC-16	FR-08	Feedback with follow-up requested	Submit rating 5 with follow-up checked	Administrator receives follow-up e-mail	pass
TC-17	FR-08	Sentiment classification	Submit comments of positive and negative tone	Sentiment field populated for each record	pass
TC-18	FR-07	Asset register grouping	Open Water Asset Register	Assets grouped by asset type	pass
TC-19	FR-09	Admin dashboard refresh	Add a complaint and reopen the Admin Dashboard	Total and pending KPIs increase by one	pass
TC-20	FR-09	Citizen sees own complaints only	Log in as citizen; open My Complaints	Only complaints with the logged-in e-mail listed	pass
TC-21	FR-09	Technician sees own details only	Log in as technician; open Technician Details	Only own technician record listed	pass
TC-22	FR-10	Daily overdue alert	Confirm schedule active; check next run at 09:00 IST	Overdue complaints e-mailed; locality PDF attached	pass

6.3 Debugging Deluge Scripts

Deluge workflows were debugged with four techniques. First, the info statement writes a value to the execution log. Second, workflows can be tested in live mode, where the saved record shows whether the status change ran. Third, the e-mail recipients can be checked by submitting a test record with known addresses. Fourth, the application's execution logs show which workflow ran, with the time and any error message.

Typical debugging steps were: confirm the field link names used in the script (for example Repair_Status rather than the display name), confirm that the trigger is the intended one (on edit of a specific field, or on success of an add), and confirm that the record variable used for the status change is the one that will be saved.

```
if(input.Issue != null)
{
issueRecord = Water_Issue_Report[ID == input.Issue];
if(issueRecord.count() > 0)
{
if(input.Verification_Status == "Approved")
{
issueRecord.Status="Resolved";
issueRecord.Resolution_Notes=input.Remarks;
sendmail
[
from :zoho.adminuserid
to :issueRecord.Citizen.Email
subject :"Water Complaint Resolved"
message :"Your water complaint has been marked Resolved after supervisor
    verification.\n\nIssue ID: " + issueRecord.Issue_ID + "\nIssue: " +
    issueRecord.Issue_Title + "\nResolution: " + input.Remarks + "\n\nYou can log in to
    the application and submit feedback."
]
}
else if(input.Verification_Status == "Rework Required")
{
issueRecord.Status="In Progress";
sendmail
[
from :zoho.adminuserid
to :input.Technician.Email
subject :"Repair Requires Further Work"
message :"The supervisor has requested rework for Repair ID: " + input.Repair_ID + ".
    Please review the inspection results and update the repair."
]
}
}
}
```

Figure 6.1: Deluge Script for Repair Verification (WF_Repair_Verification) in the Deluge Editor

6.4 Defect Log and Review Findings

A review of the exported application (export generated on 6 November 2026) against its design and against the test cases identified the points in Table 6.2. They are recorded as open items with a proposed fix. The status of each item should be confirmed during live testing before the report is submitted.

Table 6.2: Defect Log and Review Findings

ID	Finding	Component	Severity	Proposed Fix
BUG-01	The Admin and Supervisor dashboards count repairs awaiting verification with the status “Awaiting Verification”, which is not a value of the complaint status list; the KPI will show zero.	Admin Dashboard KPI	High	Count Repair Resolution records with Verification Status = Pending, as the Technician and Supervisor dashboards do
BUG-02	The Supervisor dashboard counts “Reported” complaints, a value not in the status list.	Supervisor Dashboard KPI	Medium	Use the status submitted
BUG-03	The status value “submitted” is lower case while the other values are title case.	Water Issue Report, Status	Low	Rename to Submitted, and update the workflow and dashboard criteria
BUG-04	Workflows set complaint status on a record fetched into a variable. Persistence must be confirmed in testing; if the change is not saved, an update statement is required.	WF_Repair_Progress, WF_Repair_Verification	Medium	Confirm with TC-08 to TC-12; use the update statement if needed
BUG-05	Availability values differ: Technician has Available, Busy, On Leave, Unavailable; Maintenance Team has Available, Busy, Inactive.	Technician, Maintenance Team	Low	Use a shared list of availability values
BUG-06	The overdue alert sends one e-mail per overdue complaint, which can flood the administrator’s mailbox.	WF_Daily_Overdue_Alert	Medium	Send a single digest e-mail listing all overdue complaints
BUG-07	The feedback follow-up alert runs only on add, so a later edit that requests follow-up does not alert anyone.	WF_Feedback_Follow_Up	Low	Also run the check on edit
BUG-08	The Repair Resolution form has no check that the end time is after the start time.	Repair Resolution	Low	Add a validation rule comparing the two times

ID	Finding	Component	Severity	Proposed Fix
BUG-09	Supervisor Verification Email runs on every update of Repair Status, so it e-mails "Repair Awaiting Verification" for In Progress and On Hold as well as Completed. It also duplicates the completion e-mail sent by WF_Repair_Progress.	Supervisor Verification Email, WF_Repair_Progress	Medium	Add a condition so the e-mail is sent only when Repair_Status is Completed, and keep a single completion e-mail

6.5 User Acceptance Testing

User acceptance testing is performed by asking representative users (an administrator, a supervisor, a technician and a citizen) to complete a short set of tasks: report a water issue, assign and repair it, verify the repair, submit feedback and open the dashboard. Their feedback and the action taken are recorded in Table 6.3.

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Municipal office staff	Administrator	The dashboard charts and KPI panels gave a clear picture of open, overdue and resolved complaints by area.	No change needed
Section supervisor	Supervisor	Assigning a team and technician was quick, and the verification e-mail showed the repair details. SMS alerts would help in urgent cases.	Recorded as a future enhancement (Section 9.2)
Field technician	Technician	The assigned complaints were easy to find and the repair form with materials and checklist was simple to fill in on the mobile app.	No change needed
Local resident	Citizen	The complaint form with photo upload was simple and quick. The confirmation e-mail arrived and showed the complaint status.	No change needed

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Zoho Creator provides Development, Stage and Production environments. Changes are made in Development, published to Stage for verification as a versioned release, and then published to Production, where end users work. The deployment procedure is: open Environments under Deploy, choose Publish to Stage, select the application and its changed components, enter a title and description; after verifying the Stage version, choose Publish to Production. Published versions cannot be rolled back, so the rollback plan is to publish a corrected version and to keep a .ds export of every released version as a backup. A .ds export of this application (version 1.0, 27 October 2026) was taken for that purpose.

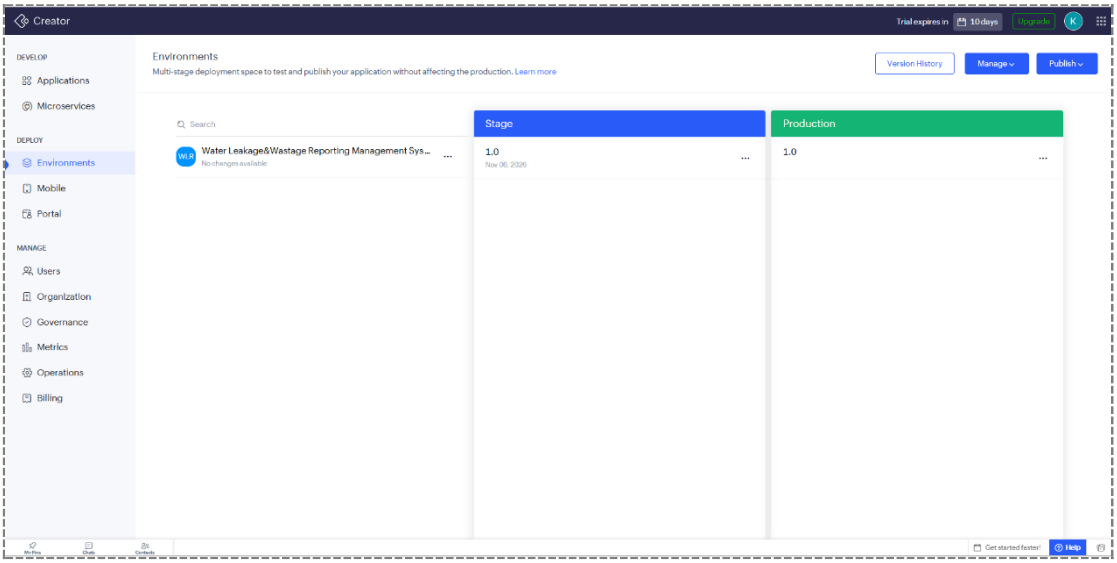


Figure 7.1: Environments – Publish to Stage and Production

Table 7.1: Deployment Checklist

S.No.	Checklist Item	✓ or ✗
1	All test cases in Table 6.1 executed in Development / Stage	✓
2	Sample and test data removed or isolated	✓
3	Profiles, roles and users configured in Production	✓
4	Workflows enabled and verified, including the scheduled overdue alert	✓
5	Notification e-mail addresses confirmed	✓
6	Application published and URL verified	✓

S.No.	Checklist Item	✓ or ✗
7	Version / release notes recorded; .ds export stored	✓ (v1.0 export)

7.2 Performance Optimisation

The application is small, so optimisation focused on keeping the work inside the platform's efficient paths. Status changes are made once in the workflow that triggers them, reports read stored values, and lookups to master forms keep records consistent. Table 7.2 lists the measures applied and the further measures recommended as data volumes grow.

Table 7.2: Performance Optimisation Measures

Measure	Component	Effect	Status
Status changes performed in the workflow that triggers them	Repair and verification workflows	No separate update step is needed	Applied
Lookups to master forms instead of free text	All transaction forms	Consistent grouping and smaller records	Applied
Report criteria so users see only their own records	Citizen and technician reports	Smaller result sets for each user	Applied
Filters on Pending Issues report	Pending Issues	Quicker review of open work	Applied
Single digest e-mail for overdue complaints	WF_Daily_Overdue_Alert	Fewer e-mails per day	Recommended (BUG-06)

7.3 Security Features

Table 7.3: Security Controls Implemented

Control	Implementation	Evidence
Authentication	Zoho account sign-in; only invited users can access the application	Platform feature
Profile-based permissions	Admin, Supervisor, Technician and Citizen profiles and roles	Section 5.8
Personal data flag	Citizen and technician name, e-mail and phone fields marked as personal data	Form definitions
Record-level access	Citizen and technician reports filtered on the logged-in e-mail	Section 5.3
Input validation	Mandatory fields, restricted drop-downs, e-mail and phone types, numeric and currency types	Tables 4.2–4.4
Transport security	Application served over HTTPS by Zoho	Platform feature
Audit trail	System fields (added user and time) and execution logs	Section 7.4

7.4 Governance and Administration

Governance covers naming, ownership, change and monitoring. Forms use consistent link names, shared logic is kept in named workflows, and every release is exported as a .ds file. The administrator owns the master data and should review the team, technician and asset lists at least once a quarter. Execution logs under Operations show workflow runs and are the first place to look when a status change or e-mail does not happen. Data stewardship rests with the department that owns each form: supervisors for complaints and repairs, technicians for repair records, and the administrator for masters and assets.

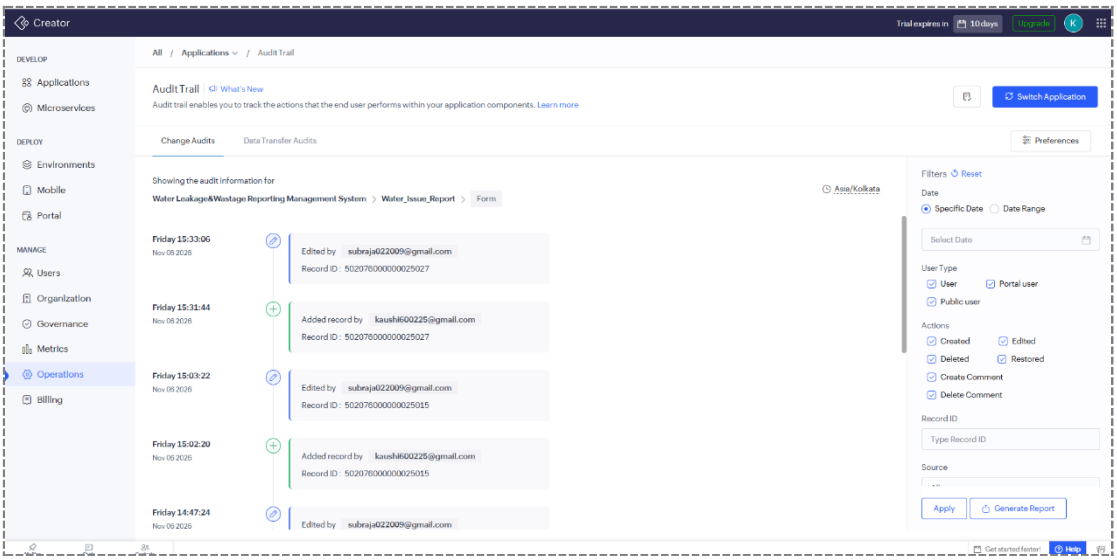
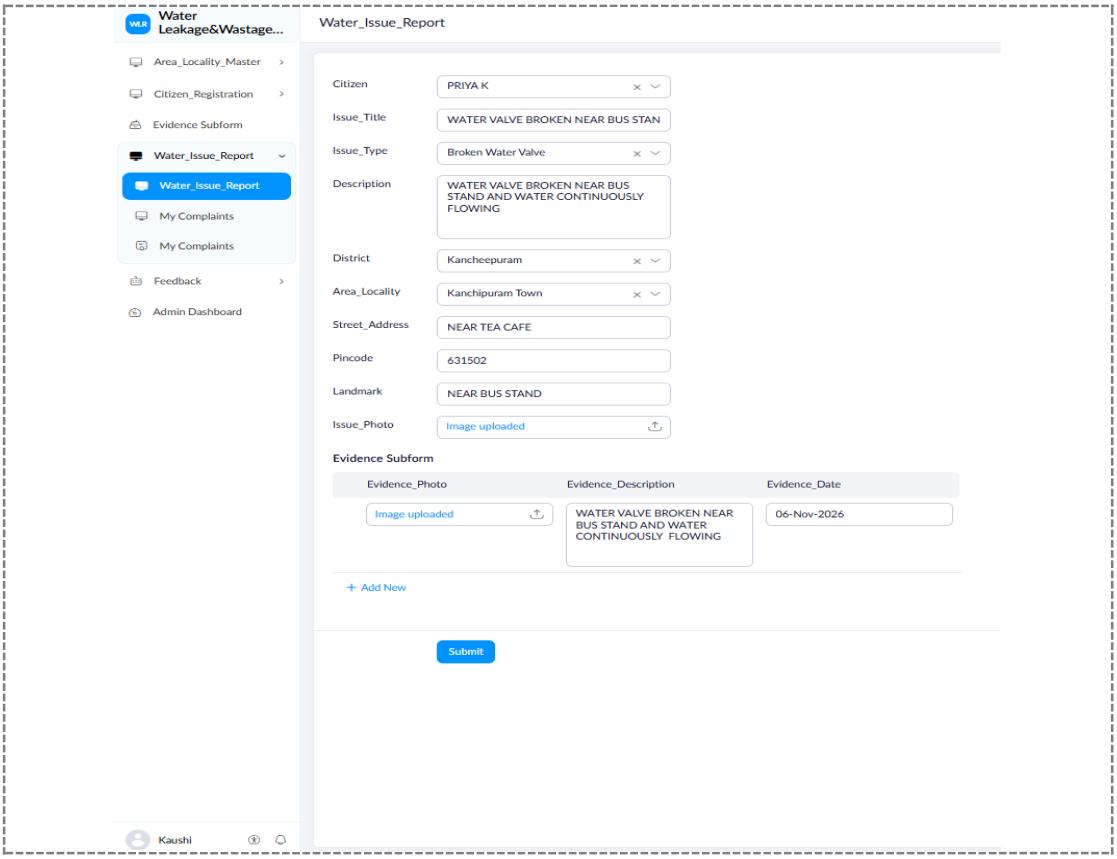


Figure 7.2: Audit trial

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The following scenario demonstrates the application end to end. The administrator creates the localities and maintenance teams and registers a technician. A citizen registers and reports a burst pipeline with a photo; the complaint is stored with status submitted and the confirmation e-mails are sent. The supervisor assigns the complaint to a team and technician, and the status becomes Assigned. The technician records a repair with materials and an inspection checklist, and marks it Completed. The supervisor verifies the repair as Approved, the complaint is resolved, and the citizen receives the resolution e-mail. The citizen then submits feedback with a rating and comments. The administrator opens the Admin Dashboard and reviews the totals and the feedback report. Screenshots of each step will be inserted at the marked positions.



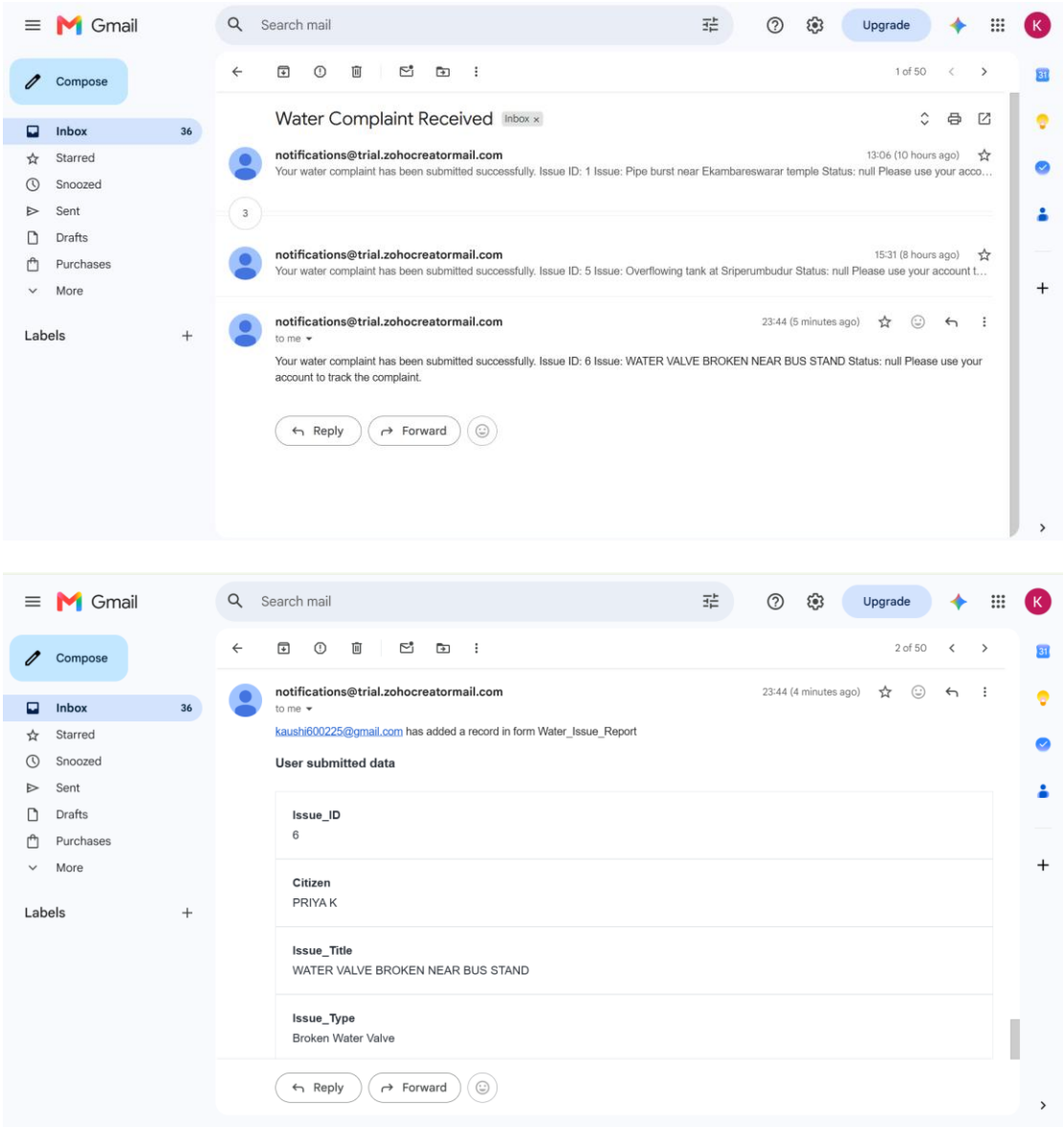


Figure 8.1: Scenario Step 1 – Complaint Submitted and Confirmed

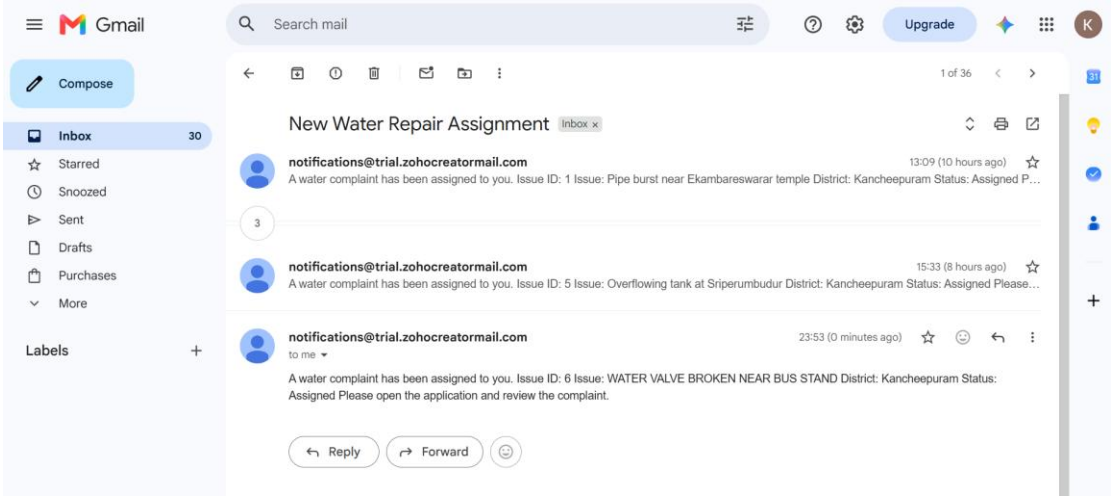


Figure 8.2: Scenario Step 2 – Technician Assignment from Supervisor

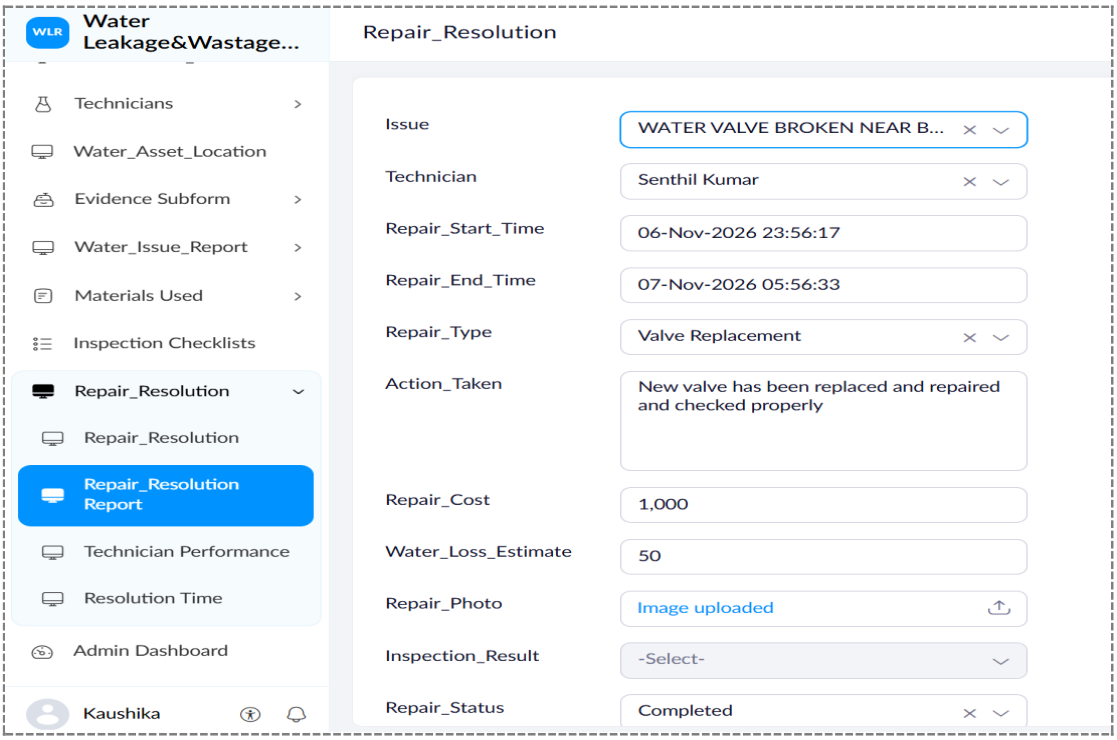


Figure 8.3: Scenario Step 3 – Repair Completed by Technician

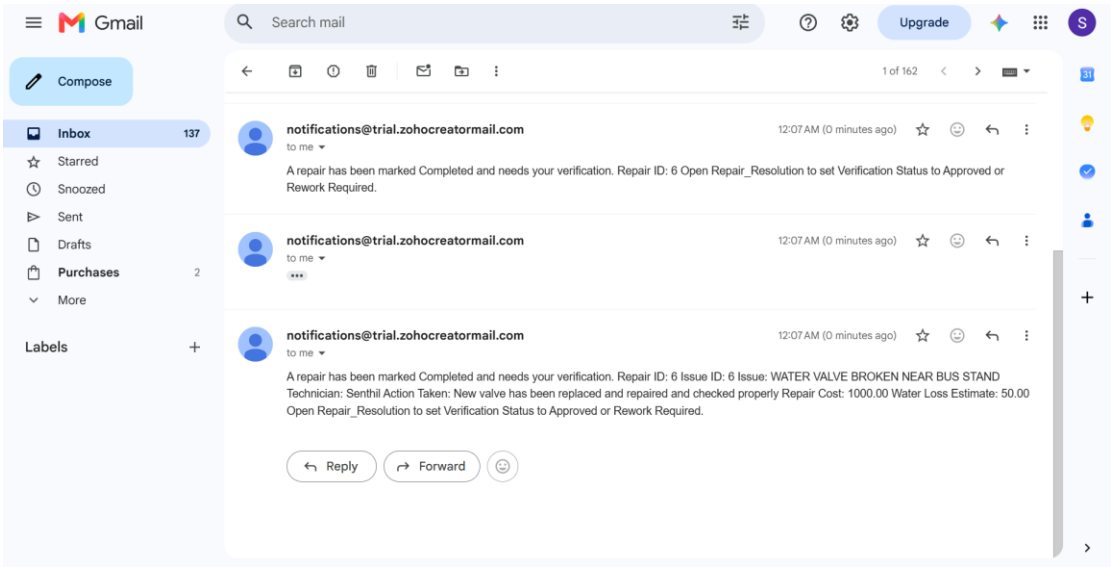


Figure 8.4: Scenario Step 4 – Completed Work Waiting for Verification by the Supervisor

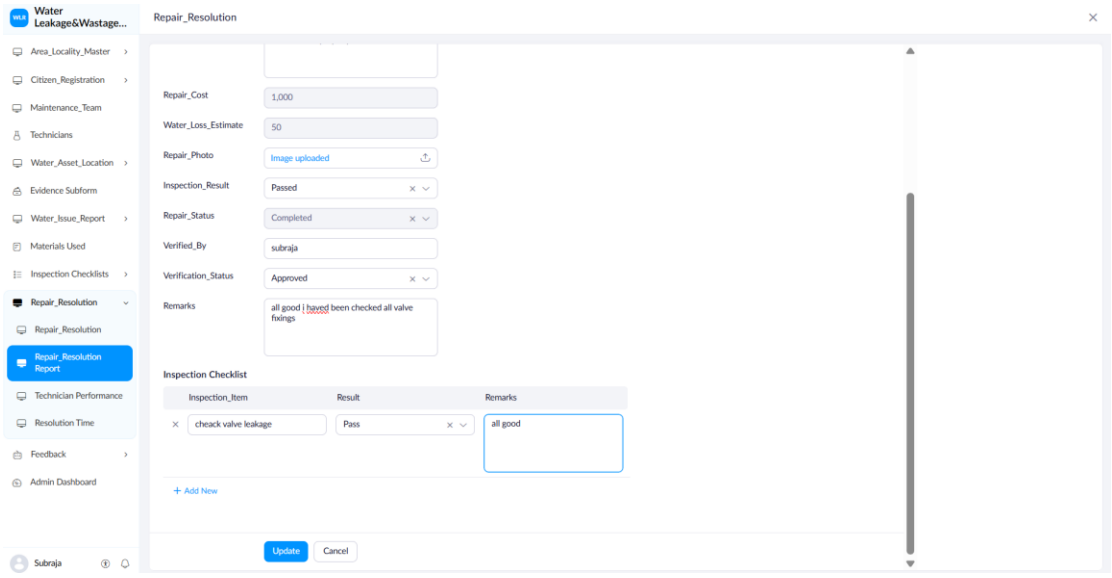


Figure 8.5: Scenario Step 5 – Verification Done by Supervisor

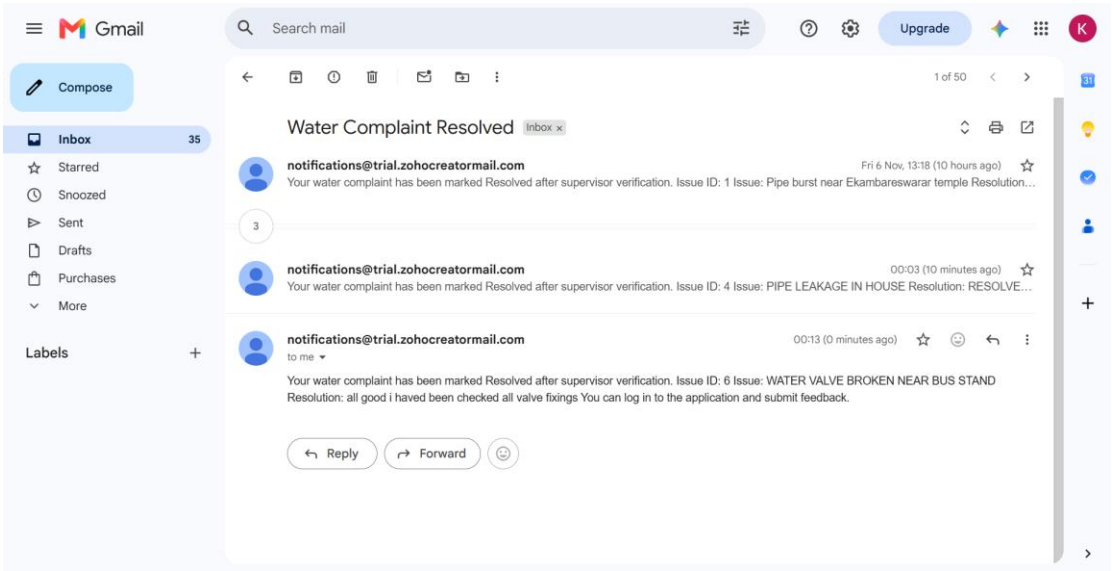


Figure 8.6: Scenario Step 6 – E-mail Sent for the Resolved Issue

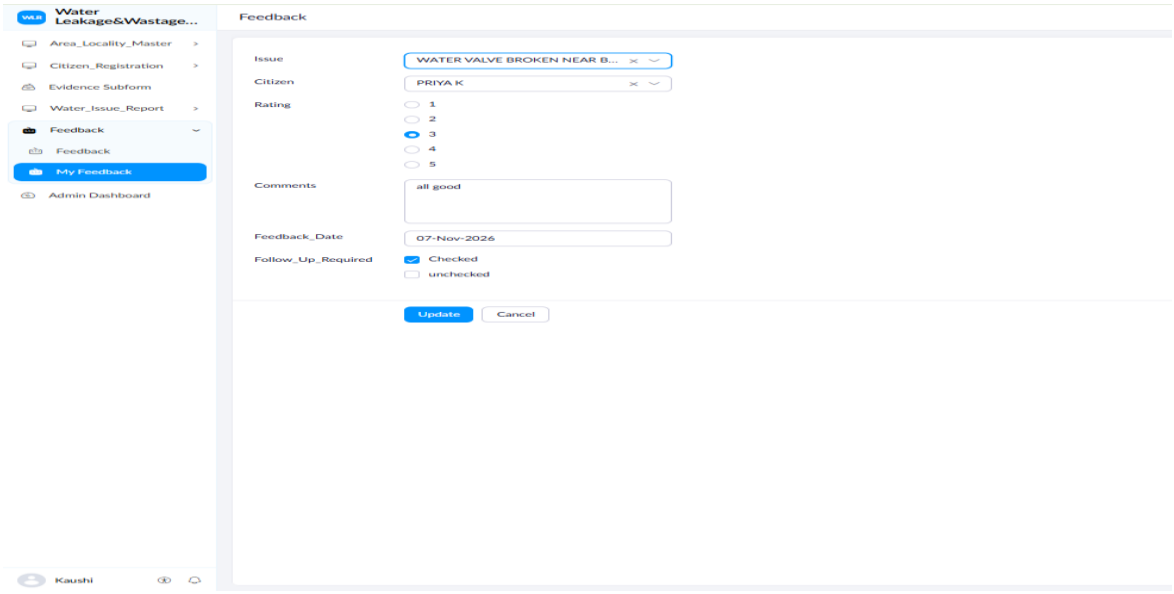
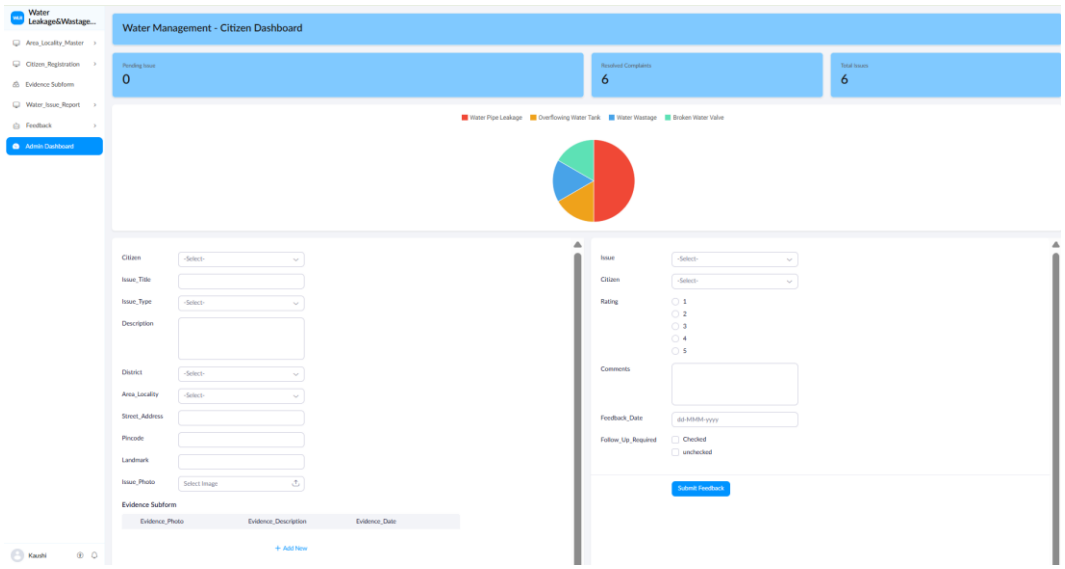


Figure 8.7: Scenario Step 7 – Feedback by Citizen



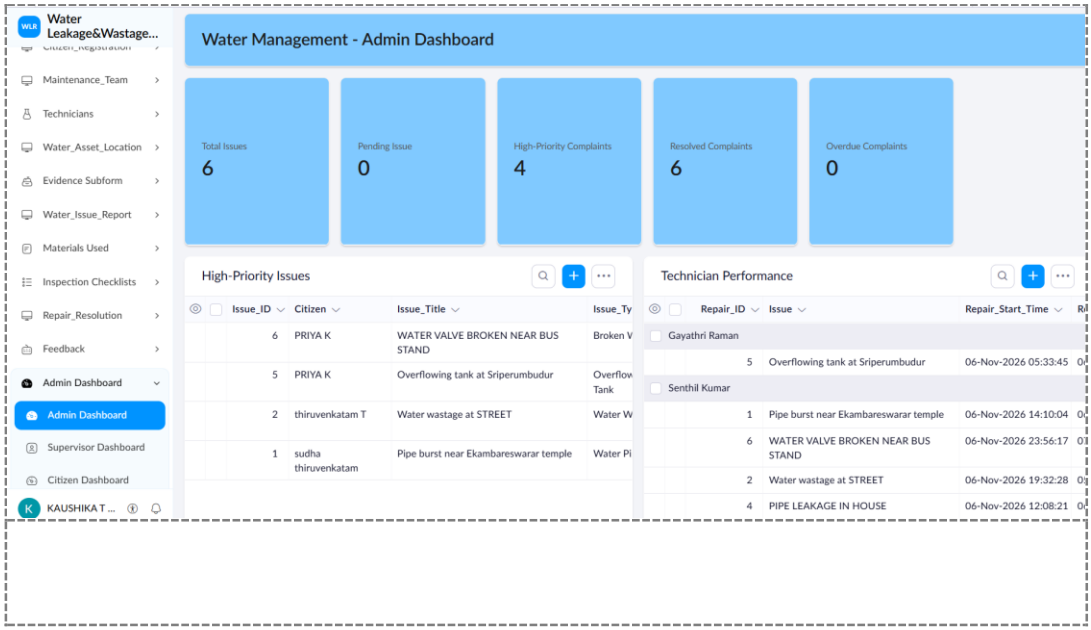


Figure 8.8: Scenario Step 8 – Dashboard Updated

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective	Outcome	Evidence
Data model and forms for all subjects	Eleven linked forms with lookups, grids and controlled drop-downs	Tables 4.1–4.4
Complaint lifecycle	Status changes through assignment, repair and verification	Workflows DS-02 to DS-04
Notifications	E-mails to citizens, technicians and supervisors at each step	DS-01, DS-02, DS-03, DS-04, DS-05, DS-07, DS-08
Reports and dashboards	27 reports and four role-based dashboards	Tables 5.1, 5.2
Scheduled alert	Daily overdue alert at 09:00 IST	DS-06
Feedback and sentiment	Feedback with follow-up alerts and sentiment field	DS-05; Section 5.9
Role separation	Four roles in a hierarchy with profiles and personal-data flags	Section 5.8, Table 4.5

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Aspect	Existing System	Proposed Application
Complaint records	Registers and messages	Single database with lookups and status
Ownership	Not recorded	Team and technician assigned; status tracked
Repair records	Verbal reports	Repair times, cost, water loss, materials and checklist recorded

Aspect	Existing System	Proposed Application
Verification	Informal	Supervisor verification with resolution notes
Overdue work	Noticed late, if at all	Daily scheduled alert
Citizen feedback	Not collected	Rating, comments and sentiment; follow-up alerts
Management view	Manual report at month end	Live dashboards and reports

8.4 Limitations of the Low-Code Approach

- The page builder offers a fixed set of chart types and styles.
- Business rules in workflows are not visible in the data model, so they must be documented, as the inconsistent status values in Table 6.2 show.
- E-mail is the only notification channel; SMS or push notifications would need an external service.
- The application depends on Zoho's hosting, subscription and release cycle.
- Water-loss figures are typed in by technicians; no flow sensor data is used.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The forms, reports and dashboards were produced without code, which is the no-code end of the spectrum. The status changes, notifications and the overdue alert needed short Deluge scripts, which is the low-code middle. The project shows that most of the application could be delivered visually, and that targeted code adds the behaviour that makes the workflow reliable.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

This project built the Water Leakage & Wastage Reporting Management System, a Zoho Creator application that brings citizen complaints, assignments, repairs, verification, assets, materials, inspections and feedback into one system. It routes each complaint through its status life cycle, notifies the people involved, runs a daily overdue alert, presents role-based dashboards and reports and analyses feedback sentiment. The project used the main capabilities of the platform covered in the course: data modelling, forms, grids, reports, pages, Deluge scripting, form and scheduled workflows, AI fields and profiles. The review recorded in Table 6.2 also shows where careful testing improves the result.

9.2 Future Enhancements

- Fix the status and dashboard inconsistencies (BUG-01 to BUG-03) and confirm the status-update behaviour in testing (BUG-04).
- Replace the overdue alert e-mail per complaint with a single digest (BUG-06).
- Add a complaint-closure approval process or Blueprint so that closing a complaint needs a second approval.
- Send SMS or push notifications for assignments and resolutions through an external service.
- Add trend comparisons (month on month) and ranking of localities by water loss.
- Integrate water meters or flow sensors so that readings flow into the application automatically.
- Add a leaderboard for technicians based on repairs verified on time.
- Add custom permission sets for each district so that each office sees only its own complaints.

9.3 Learning Outcomes and Career Relevance

The project developed skills in modelling data with forms, lookups and grids; writing and debugging Deluge workflows; building role-based dashboards; designing profiles

and access; and reviewing an application critically against its design. These skills map to roles such as citizen developer, low-code developer, Zoho Creator consultant and automation engineer, and the experience of extending a platform with code is a step towards professional software development.

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[1] A. Sahay, A. Indamutsa, D. Di Ruscio and A. Pierantonio, “Supporting the understanding and comparison of low-code development platforms,” in Proc. 46th Euromicro Conf. Software Engineering and Advanced Applications (SEAA), 2020.

[2] A. C. Bock and U. Frank, “Low-code platform,” Business & Information Systems Engineering, vol. 63, no. 6, pp. 733–740, 2021.

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[4] Zoho Corporation, “Zoho Creator Help.” [Online]. Available: https://www.zoho.com/creator/help/

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section
I	Application created and sections organised	Y	Navigation sections	5.1
II	Forms with at least eight field types	Y	Text, number, date, date-time, e-mail, phone, address, image, drop-down, lookup, currency, AI	5.2
II	Lookup field (relationship between forms)	Y	Area, Citizen, Team, Technician, Issue lookups	4.2
II	Subform (parent–child data)	Y	Evidence, Materials Used, Inspection Checklist grids	4.2
II	Validations and layout rules	Y	Mandatory fields, drop-downs, field types	5.2
II	List report with filter, grouping	Y	Pending Issues filters; grouped reports	5.3
II	Kanban / Calendar / Timeline / Map report	N	Not configured	9.2
II	Pivot table / chart report	Y	Issue Type Split pivot chart on Water Issue Report	5.3
II	Page / dashboard built with page builder	Y	Four role dashboards	5.4
II	Deluge – form events (validate / success / edit)	Y	Six form workflows	5.5
II	Deluge – functions	N	No custom functions	—
II	E-mail notifications	Y	sendmail in workflows	5.7
II	Form workflow	Y	Six form workflows	5.6
II	Scheduled workflow	Y	WF_Daily_Overdue_Alert	5.6
II	Approval process / Blueprint / Batch workflow	Partly	Batch workflow Daily_Assignment_and_Verification_Report; approval and blueprint not used	5.6
II	Testing and debugging evidence	Y	22 planned test cases; 8 review findings	6
III	Integration with a Zoho service	Partly	No CRM and E-mail notifications (sendmail); AI sentiment-analysis field	5.7
III	Connection / REST API call	N	Not used	—
III	Users, roles and permission profiles	Y	Admin, Supervisor, Technician, Citizen profiles and roles	5.8
III	Record-level access criteria	Y	Citizen and technician reports filtered by login	5.3
III	AI feature	Y	Sentiment field on Feedback	5.9

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section
III	Analytics (charts)	Y	Dashboard charts and KPI panels	5.4
IV	Performance optimisation	Y	Single-point status updates; criteria on reports	7.2
IV	Deployment across environments	Y	Procedure and .ds export	7.1
IV	Security and governance	Y	Profiles, personal-data flags, logs	7.3, 7.4
V	Pro-code extension	N	Deluge scripts in workflows only; no external REST API, custom widget or separate code base (see 5.10)	5.5, 5.10
V	Evaluation of low-code limits	Y	Limitations and reflection	8.4, 8.5

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture and business value of low-code application development using Zoho Creator.	Ch. 1, 2		
CO2	Develop business applications using Zoho Creator forms, reports, workflows and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6		
CO3	Integrate Zoho Creator applications using notifications and role-based security.	Ch. 4.6, 5.7–5.9		
CO4	Deploy, secure and manage Zoho Creator applications across environments following governance practices.	Ch. 7		
CO5	Evaluate low-code strategies, AI-assisted features and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3		

APPENDIX C – DELUGE SOURCE CODE (EXCERPTS)

The scripts below are reproduced from the exported application (export generated on 6 November 2026). The complete scripts are in the .ds export and should be pasted from there to avoid transcription errors.

C.1: WF_Repair_Verification (form workflow, on edit of Verification Status, excerpt)

```
if(input.Issue != null)
{
 issueRecord = Water_Issue_Report[ID == input.Issue];
 if(issueRecord.count() > 0)
 {
  if(input.Verification_Status == "Approved")
  {
   issueRecord.Status = "Resolved";
   issueRecord.Resolution_Notes = input.Remarks;
   sendmail [ from: zoho.adminuserid  to: issueRecord.Citizen.Email
subject: "Water Complaint Resolved"  message: "..." ]
  }
  else if(input.Verification_Status == "Rework Required")
  {
   issueRecord.Status = "In Progress";
   sendmail [ from: zoho.adminuserid  to: input.Technician.Email
subject: "Repair Requires Further Work"  message: "..." ]
  }
 }
}
```

C.2: WF_Feedback_Follow_Up (form workflow, on success of add, excerpt)

```
if((input.Follow_Up_Required != null &&
input.Follow_Up_Required.contains("Checked")) || input.Rating == "1"
|| input.Rating == "2")
{
 sendmail [ from: zoho.adminuserid  to: zoho.adminuserid  subject:
"Citizen Feedback Requires Follow-up"  message: "..." ]
}
```

C.3: WF_Daily_Overdue_Alert (scheduled workflow, excerpt)

```
for each issue in Water_Issue_Report[Expected_Resolution_Date <
zoho.currentdate && Status != "Resolved" && Status != "Closed" &&
Status != "Rejected"]
{
 sendmail [ from: zoho.adminuserid  to: zoho.adminuserid  subject:
"Overdue Water Complaint"  message: "..." ]
}
```

C.4: WF_Repair_Progress (form workflow, on edit of Repair Status, export of 6 November 2026)

```
if(input.Issue != null)
{
complaint = Water_Issue_Report[ID == input.Issue];
if(complaint.count() > 0)
```

```
{
if(input.Repair_Status == "In Progress")
{
complaint.Status="In Progress";
}
else if(input.Repair_Status == "On Hold")
{
complaint.Status="On Hold";
}
else if(input.Repair_Status == "Completed")
{
complaint.Status="Under Review";
supervisorEmail = zoho.adminuserid;
if(complaint.Assigned_Team != null)
{
team = Maintenance_Team[ID == complaint.Assigned_Team];
if(team.count() > 0 && team.Supervisor_Email != null)
{
supervisorEmail = team.Supervisor_Email;
}
}
technicianName = "Not assigned";
if(input.Technician != null)
{
technicianName = input.Technician.Technician_Name.first_name;
}
sendmail
[
from :zoho.adminuserid
to :supervisorEmail
subject :"Repair Awaiting Verification"
message :"A repair has been marked Completed and needs your
verification.\n\nRepair ID: " + input.Repair_ID + "\nIssue ID: " +
complaint.Issue_ID + "\nIssue: " + complaint.Issue_Title +
"\nTechnician: " + technicianName + "\nAction Taken: " +
ifnull(input.Action_Taken,"") + "\nRepair Cost: " +
ifnull(input.Repair_Cost,"") + "\nWater Loss Estimate: " +
ifnull(input.Water_Loss_Estimate,"") + "\n\nOpen Repair_Resolution to
set Verification Status to Approved or Rework Required."
]
}
}
}
```

C.5: Supervisor Verification Email (form workflow, on edit of Repair Status)

```
supervisorEmail = zoho.adminuserid;
tech = Technician[ID == input.Technician];
if(tech.count() > 0 && tech.Team != null)
{
team = Maintenance_Team[ID == tech.Team];
if(team.count() > 0 && team.Supervisor_Email != null)
{
supervisorEmail = team.Supervisor_Email;
}
}
sendmail
[
from :zoho.adminuserid
to :supervisorEmail
subject :"Repair Awaiting Verification"
```

```
message :"A repair has been marked Completed and needs your
verification.\n\nRepair ID: " + input.Repair_ID + "\n\nOpen
Repair_Resolution to set Verification Status to Approved or Rework
Required."
]
```

C.6: Daily_Assignment_and_Verification_Report (batch workflow, daily)

```
unassigned = "";
for each  c in Water_Issue_Report[Technician == null && Status !=
"Resolved" && Status != "Closed" && Status != "Rejected"]
{
unassigned = unassigned + c.Issue_ID + " | " + c.Issue_Title + " | "
+ c.District + " | Status: " + c.Status + "\n";
}
if(unassigned == "")
{
unassigned = "None\n";
}
unverified = "";
for each  r in Repair_Resolution[Repair_Status == "Completed" &&
Verification_Status == "Pending"]
{
unverified = unverified + "Repair " + r.Repair_ID + " | Issue " +
r.Issue.Issue_ID + " | " + r.Issue.Issue_Title + "\n";
}
if(unverified == "")
{
unverified = "None\n";
}
sendmail
[
from :zoho.adminuserid
to :zoho.adminuserid
subject :"Daily Report: Unassigned Complaints and Unverified Repairs"
message :"UNASSIGNED COMPLAINTS (no technician assigned):\n" +
unassigned + "\n\nUNVERIFIED REPAIRS (Completed, waiting for
supervisor verification):\n" + unverified
]
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

- Accept the invitation e-mail from Zoho Creator and sign in with the same e-mail address.
- On the web, open creator.zoho.in and open Water Leakage & Wastage Reporting Management System.
- On a phone, install the Zoho Creator app, sign in with the same account and open the application.

D.2 Instructions for Administrators and Supervisors

- Set up masters first: Area/Locality Master, Maintenance Team and Technician records.
- Review new complaints in Water Issues > Water Issue Report and the Pending Issues report.
- Assign a team and technician to a complaint by editing the complaint; the technician is e-mailed.
- Verify completed repairs in Repair Resolution: set Verification Status to Approved (resolves the complaint) or Rework Required.
- Review the dashboard for the summary, and the Citizen Feedback report for follow-up.

D.3 Instructions for Technicians

- Open Repair Resolution and create a record for each assigned complaint.
- Enter start and end times, action taken, cost, water-loss estimate and photos; add materials and checklist rows.
- Set Repair Status to In Progress, On Hold or Completed.

D.4 Instructions for Citizens

- Report a problem: Water Issues > Water Issue Report. Choose the type, severity, locality and describe the issue; add photos.
- Track complaints in My Complaints, which shows status and resolution notes.
- Submit feedback after a complaint is resolved, with a rating and comments.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Table E.0: Student Details

Student Name	Register No.	Project Title	Date
T KAUSHIKA	240701239	Water Leakage & Wastage Reporting Management System	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form and scheduled workflows	15	
5	Integration, RBAC & AI	Notifications, roles & permissions, AI/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls	10	
7	Live Demonstration	End-to-end working across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		100	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation

Level	% of Max. Marks	Descriptor
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student

Signature of Internal Examiner

Signature of External Examiner

Name: _____

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Table F.0: Viva Details

Student Name	Register No.	Department / Section	Date of Viva
T KAUSHIKA	240701239	CSE / 3rd Year, Sem V	

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	100	

Examiners’ Remarks:

_____

INTERNAL EXAMINER EXTERNAL EXAMINER

Name: _____ Name: _____